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Analysis and Design of Advanced Internal Model Control Schemes for Linear Systems

TESI DI LAUREA MAGISTRALE IN
AUTOMATION AND CONTROL ENGINEERING - INGEGNERIA
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Abstract

According to the internal model principle, robust regulation for generic linear time-invariant systems can be achieved only if the controller embeds a model of the exosystem that generates the exogenous signals. As long as stability is maintained, the regulation error asymptotically vanishes under model uncertainty. This thesis formulates designs for compensation of internal models (CIM) to minimise stabiliser complexity and avoid re-design when internal model parameters are subject to variation over time. The technique is initially introduced for state-feedback systems and subsequently extended to output-feedback topologies, wherein only measured outputs are available and only a subset of outputs is subject to regulation. Under certain assumptions on the plant model, the closed-loop dynamics is affinely affected by internal model parameters, facilitating the simple online adaptation of the internal model while maintaining the stabiliser's independence. The proposed framework is validated on a grid-connected inverter benchmark. Proportional-resonant and repetitive controllers are modified by the compensation of the internal model for sinusoidal current tracking and harmonic disturbance rejection, decreasing stabiliser complexity. Simulation outcomes demonstrate reduced steady-state tracking error and decreased current harmonic distortion. Adaptive frequency tuning is examined in the context of switching grid frequency.

Keywords: output regulation; internal model principle; internal model compensation; adaptive control; output feedback; proportional-resonant control; repetitive control; grid-connected inverter; harmonic rejection

Abstract in lingua italiana

Secondo il principio del modello interno, la regolazione robusta per sistemi lineari tempo-invarianti può essere ottenuta solo se il controllore incorpora un modello dell'esosistema che genera i segnali esogeni. Finché la stabilità è mantenuta, l'errore di regolazione si annulla asintoticamente anche in presenza di incertezze di modello. Questa tesi presenta un approccio di compensazione a modello interno (CIM) per minimizzare la complessità dello stabilizzatore ed evitare riprogettazioni quando i parametri del modello interno variano nel tempo. Inizialmente, l'approccio è presentato per sistemi in retroazione di stato e successivamente esteso a topologie in retroazione d'uscita, nelle quali sono disponibili solo le uscite misurate e solo un sottoinsieme delle uscite è soggetto a regolazione. Sotto alcune ipotesi sul modello, le dinamiche in anello chiuso dipendono in modo affine dai parametri del modello interno, facilitando un semplice adattamento online di tali parametri, mantenendo l'indipendenza dello stabilizzatore rispetto ad essi. I controllori proporzionale-risonante e ripetitivi sono modificati tramite compensazione del modello interno per l'inseguimento e il rigetto di disturbi armonici, riducendo la complessità dello stabilizzatore. I risultati di simulazione mostrano una riduzione dell'errore di inseguimento a regime e una diminuzione della distorsione armonica della corrente. La taratura adattativa della frequenza viene analizzata nel contesto di variazioni della frequenza di rete dovute alla commutazione.

Parole chiave: regolazione in uscita; principio del modello interno; compensazione a modello interno; controllo adattativo; retroazione d'uscita; controllo proporzionale-risonante; controllo ripetitivo; inverter connesso alla rete; rigetto delle armoniche

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Introduction

In many control applications, the objective is not only to stabilise a plant but also to achieve asymptotic tracking of reference signals and rejection of persistent disturbances. Typical exogenous inputs include constant components, harmonics, their combinations, and general periodic signals. Classical linear feedback has shown that error feedback can minimise steady-state error; however, this comes at the cost of a rather high loop gain. The concept of including an internal model of the signals to be rejected or tracked has shown that a zero steady-state error could be attained with a low gain.

A central theoretical result addressing this class of problems is the Internal Model Principle (IMP) [12]. In its traditional linear framework, the IMP claims that effective regulation is attained if and only if the controller incorporates an appropriate representation of the exosystem producing the reference/disturbance signals. The closed-loop transfer function from external inputs to the regulatory error must possess transmission zeros that include the eigenvalues of the exosystem. This condition clarifies the necessity of integral action for the rejection of constant disturbances and the imperative of resonant dynamics for the rejection or tracking of sinusoidal signals. The IMP serves not only as a design guideline but also as a declaration of robustness: once the internal model is established and closed-loop stability is ensured, the asymptotic reduction of the regulation error can be sustained despite uncertainties in the plant [22].

Although the idea is conceptually straightforward, its application can be difficult in practical designs. The inclusion of the internal model in the control loop may significantly increase the controller order, particularly in cases of repetitive control [15], and can complicate stabilisation when the plant is only partially observed or when only a subset of the outputs requires regulation. Furthermore, robustness criteria are complex: the internal model must be appropriately linked to each regulated output channel with the appropriate geometric multiplicity, and the interconnection must be stabilisable. These factors require systematic synthesis methods that maintain the advantages of the IMP while managing complexity and ensuring implementability.

This thesis examines these challenges by analysing and expanding a category of control

strategies centred on internal model compensation. The fundamental concept is to distinguish the function of the internal model (which guarantees the appropriate steady-state configuration for regulation) from the function of the stabiliser (which ensures closed-loop stability and influences transient performance) [10]. In combination with a canonical parametrisation of a linear exogenous system [30], it facilitates the implementation of adaptation laws that achieve asymptotic regulation or, under appropriate conditions, asymptotic estimates of the unknown exosystem parameters.

Another motivation concerns grid-connected inverters. In such systems, high power-quality standards require accurate tracking of sinusoidal references and the elimination of harmonic disturbances to minimise the total harmonic distortion (THD) of the injected current. Proportional-resonant (PR) and repetitive control are common approaches [18] as they implement a model for sinusoidal and periodic signals by generating high loop gain at desired frequencies. Nevertheless, real constraints emerge from plant uncertainty, as well as delays caused by sampling and fluctuations in grid conditions (such as variations in grid frequency and voltage harmonic distortion). These effects constrain the attainable bandwidth and the amount of harmonics that may be suppressed while maintaining sufficient stability margins. Consequently, there exists a strong incentive to create systematic internal-model-based controllers that maintain robustness in the presence of delays and uncertainty, and to enhance them with adaptation when the parameters of external signals are not precisely known.

The primary focus of this thesis is a linear time-invariant system that is affected by continuously persistent, but bounded, external signals such as references and disturbances originating from an exosystem. This exosystem is typically defined by modes that lie on the imaginary axis and may include parameters that change over time. The primary objectives are to:

1. develop stabilising controllers that ensure asymptotic regulation by incorporating a suitable internal model while managing the complexity of the stabiliser;
2. enhance the design to cover output regulation where only a subset of outputs is controlled and complete state measurement is unavailable;
3. implement adaptation laws for the parameters of the internal model;
4. assess the efficiency of the proposed design methodology on a grid-connected inverter benchmark, emphasising current regulation and harmonic rejection, and assess performance in terms of tracking error and total harmonic distortion (THD).

Thesis outline

This thesis is organised as follows.

- Chapter 1 addresses the core Internal Model Principle for SISO systems and shows how to implement internal model compensation ideas in a state-feedback framework. It also provides an example of persistent disturbance rejection and adaptation to its parameter change.
- Chapter 2 broadens the concept of internal model compensation to encompass the output-feedback context, establishes the necessary controller structures for adaptation, and introduces an algorithm for adapting internal model parameters that is appropriate for output-feedback applications.
- Chapter 3 presents the grid-connected inverter benchmark, describes plant modelling, and examines limitations such as delay-induced phase lag and its effects on harmonic rejection capability.
- Chapter 4 discusses the suggested inverter control schemes that utilise internal model compensation. This encompasses both proportional-resonant and repetitive control designs. Additionally, the chapter examines adaptation techniques capable of addressing frequency mismatches.
- Chapter 5 concludes the thesis and proposes potential areas for future research.

1 | Compensation of internal model for state-feedback regulation

1.1. Classical Internal Model Principle

The Internal Model Principle was proposed for linear time-invariant systems by Francis and Wonham [12]. The principle, in its original formulation, states that precise asymptotic regulation is possible only if the controller incorporates an internal model of the exosystem that produces the reference or disturbance signals unless the plant does not possess it from the very beginning.

In the linear scenario, this requirement necessitates the incorporation of the disturbance characteristic polynomial into the controller dynamics. Given appropriate coprimeness and stability requirements, this ensures complete asymptotic rejection of persistent disturbances.

In contrast to existing control strategies, such as high-gain or sliding-mode, which attain disturbance attenuation at the expense of chattering and sensitivity to measurement noise, the Internal Model Principle facilitates precise asymptotic disturbance rejection with reduced cost. Precise asymptotic disturbance rejection is attained by producing infinite loop gain selectively at the relevant frequencies, rather than generally increasing the feedback gain.

1.1.1. Motivating example: persistent disturbance rejection

A simple and clear illustration of this principle is provided by integral control in the presence of constant disturbances. Consider a constant disturbance $d(t)$ that generates

an additive effect on the plant input. The Laplace transform is given by

$$D(s) = \frac{\bar{d}}{s}$$

Consequently, constant disturbances are defined by a pole at the origin.

A PID controller is characterised by the equation

$$PID(s) = k_P + k_I \frac{1}{s} + k_D s$$

which incorporates an integrator, representing an internal model of constant signals. The alignment of poles between the disturbance and the controller is the fundamental mechanism of disturbance rejection.

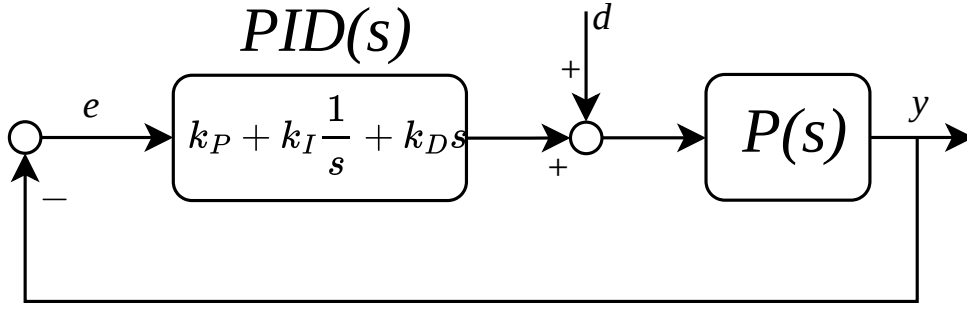


Figure 1.1: Feedback system with additive disturbance and internal model in the controller

Consider the first-order plant (see Figure 1.1)

$$G(s) = \frac{1}{ms + b},$$

subject to an additive constant disturbance $d(t) = \bar{d} \text{step}(t)$ at the input. The closed-loop error in the Laplace domain is

$$E(s) = -Y(s) = -G_{d \rightarrow y}(s) \frac{\bar{d}}{s} = -\frac{G(s)}{1 + G(s)PID(s)} \frac{\bar{d}}{s}$$

Substituting $G(s)$ and $PID(s)$ yields

$$\begin{aligned} E(s) &= -\frac{\frac{1}{ms+b}}{1 + \frac{1}{ms+b} (k_P + k_I \frac{1}{s} + k_D s)} \frac{\bar{d}}{s} = -\frac{1}{ms + b + k_P + k_I \frac{1}{s} + k_D s} \frac{\bar{d}}{s} \\ &= -\frac{s}{ms^2 + bs + k_P s + k_I + k_D s^2} \frac{\bar{d}}{s}. \end{aligned}$$

The factor s associated with the disturbance dynamics is cancelled by the integrator in the controller, leading to

$$E(s) = -\frac{\bar{d}}{ms^2 + bs + k_P s + k_I + k_D s^2}$$

If the closed-loop characteristic polynomial is Hurwitz (i.e., all its roots lie in the open left half-plane), then

$$\lim_{t \rightarrow \infty} e(t) = 0,$$

independently of the value of $\bar{d}$.

This simple example shows that constant disturbances are asymptotically rejected because the controller embeds their generating dynamics.

1.1.2. General formulation of the Internal Model Principle

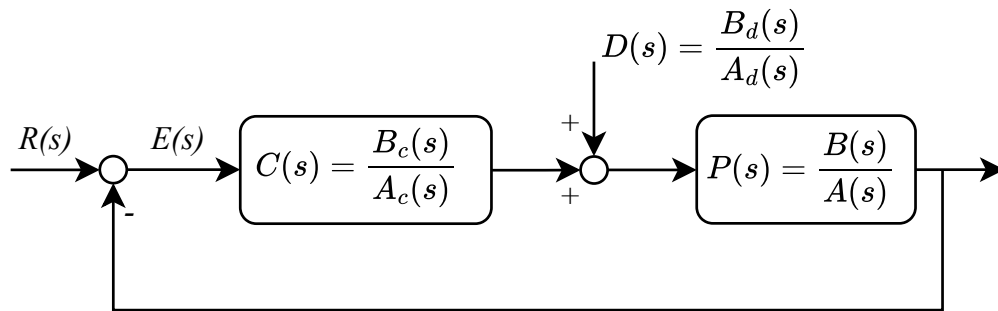


Figure 1.2: General disturbance rejection framework and internal model structure

Consider a linear time-invariant plant (see Figure 1.2)

$$P(s) = \frac{B(s)}{A(s)},$$

subject to a disturbance generated by

$$D(s) = \frac{B_d(s)}{A_d(s)}.$$

Let the controller be

$$C(s) = \frac{B_c(s)}{A_c(s)}.$$

Assume that

- $A(s)$ and $B(s)$ are coprime,
- $A_d(s)$ and $B(s)$ are coprime,
- the closed-loop system is internally stable, i.e., the polynomial $B(s)B_c(s) + A(s)A_c(s)$ is Hurwitz,
- the controller denominator satisfies

$$A_c(s) = A_d(s)A'_c(s),$$

where $A'_c(s)$ can be selected to shape closed-loop stability and performance.

Then the disturbance generated by $D(s)$ is asymptotically rejected.

In fact, the internal model principle is a robustness necessity: asymptotic regulation under plant uncertainty requires the closed-loop map from the exogenous signal to the regulation error to have zeros at the exosystem poles; if the plant does not already provide these zeros, they must be introduced by the controller; i.e., the controller must embed the exosystem model (up to stable factors).

This result formalises the intuitive idea illustrated by the PID example: exact disturbance rejection is achieved only if the controller contains an internal copy of the disturbance dynamics.

1.1.3. Limitations of the classical formulation

The first limitation occurs when an internal model is embedded in a controller; in order to create a stabiliser with the necessary performance, both the internal model and the plant must be taken into account. In situations like repetitive control, when the model is infinite-dimensional, the complexity may rise dramatically [29].

Second, the external signal may drift or switch in real-world problems, whereas IMP

only rejects fixed modes. In this case, altering the parameter requires redesigning the stabiliser, which could make adaptive control implementation more difficult. Furthermore, there may be a complicated and nonlinear dependence on the model's parameters (like modes), necessitating the use of computationally demanding adaptation techniques and recalculation of the controller parameters.

The recent work [9, 19] aimed to design a controller based on the internal model principle with compensators so that the stabiliser can be separated from the internal model. Moreover, the controller structure was modified so that the parameters of the model appeared affine in the controller, making it possible to apply the adaptation. The controller scheme is designed for both state [8] and output feedback [20].

1.2. CIM for state feedback regulation

In this section, the discussion moves from the transfer-function formulation of the internal model principle to a state-space, state-feedback setting, where the compensation of the internal model (CIM) construction is introduced.

Consider the plant

$$\dot{x}(t) = Ax(t) + Bd(t) + Bu(t), \quad (1.1)$$

where $x(t) \in \mathbb{R}^n$ is its measurable state, $u \in \mathbb{R}^m$ is a control input, and $d \in \mathbb{R}^m$ is a load disturbance. B has full column rank, and (A, B) is stabilisable. Assume that the whole state is measured and that the disturbance can be represented in the form

$$D(s) = M(s)D_0(s), \quad (1.2)$$

in the Laplace domain, where $M(s)$ is a known model (exosystem) and $D_0(s) \in H_\infty$, meaning that $D_0(s)$ is proper with all poles having negative real parts.

Assumption 1.1.

$$M(\infty) = I \quad \text{and} \quad M^{-1} \in H_\infty.$$

This assumption is not restrictive (see [8]). From [19] the control law is

$$u = M(v + R_{stab}x), \quad (1.3)$$

where M is incorporated into the controller to asymptotically reject the disturbance d

by the Internal Model Principle, $v \in \mathbb{R}^m$ can be introduced as a reference signal. The stabiliser is defined as

$$R_{stab} = \bar{R} - \Pi, \quad (1.4)$$

$\Pi \in H_\infty$ is any (stable) system such that

$$\Pi(s)(sI - A)^{-1}B = I - M^{-1}(s). \quad (1.5)$$

The possible choice of $\bar{R}$ is

$$\bar{R} = K,$$

where $K \in \mathbb{R}^{m \times n}$ is such that $A + BK$ is Hurwitz.

In the following derivations, the Laplace domain is used; dependence on s and on t is omitted when clear from context. The condition (1.5) arises from the next manipulations. The state x from (1.1) is

$$x = (sI - A)^{-1}B(d + u).$$

Plugging it into the control law (1.3) one gets

$$u = M(v + Kx - \Pi(sI - A)^{-1}B(d + u)). \quad (1.6)$$

With (1.5) in (1.6):

$$\begin{aligned} u &= M(v + Kx - (I - M^{-1})(d + u)) \\ u &= Mv + MKx - Md + MM^{-1}d - Mu + MM^{-1}u \\ u &= Mv + MKx - Md + d - Mu + u \\ Mu &= Mv + MKx - Md + d \end{aligned}$$

Finally,

$$u = v + Kx - d + M^{-1}d. \quad (1.7)$$

When it is substituted into the plant (1.1), a closed-loop system is obtained

$$\dot{x}(t) = (A + BK)x(t) + Bd(t) + Bv(t) - Bd(t) + Bd_{eq}(t),$$

where $d_{eq} = M^{-1}d$. A closed-loop dynamics then yields

$$\dot{x}(t) = (A + BK)x(t) + Bv(t) + Bd_{eq}(t). \quad (1.8)$$

Taking into account (1.2) $d_{eq} = d_0$ and decays exponentially. It is possible that d_{eq} is more complex and even non-decaying, but if M contains unstable modes of d they will be filtered out.

While the distinction between the internal model and the external control signal v is evident from (1.8), it remains beneficial to analyse the configurations of closed-loop systems for the sole stabilisation of the plant (Figure 1.3a) as well as stabilisation incorporating the internal model and its compensator (Figure 1.3b).

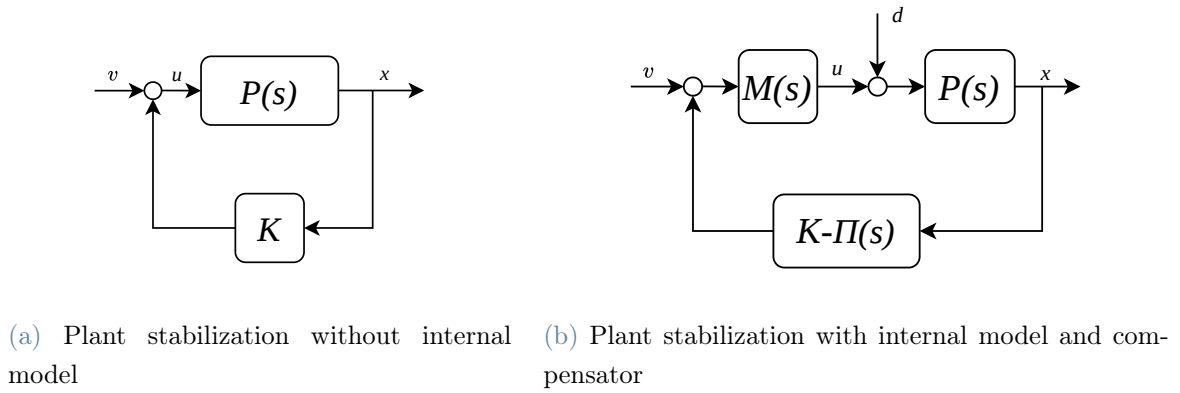


Figure 1.3: Comparison of closed-loop configurations.

The following notation is introduced.

Notation 1.1. Given a full column rank matrix $M \in \mathbb{R}^{n \times m}$, its left Moore-Penrose pseudoinverse is denoted as $M^\# \in \mathbb{R}^{m \times n}$, and defined as

$$M^\# = (M' M)^{-1} M',$$

so that $M^\# M = I$.

The compact notation is used

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = D + C(sI - A)^{-1} B$$

for transfer functions in terms of their state-space realizations.

With this notation the plant (1.1) can be written as

$$P(s) = \left[\begin{array}{c|c} A & B \\ \hline I & 0 \end{array} \right]. \quad (1.9)$$

Notice that in (1.5) $(sI - A)^{-1}B = P$ and introduce its left inverse, yielding

$$P^\# P = I,$$

where $P^\#$ is

$$P^\# = B^\# s - B^\# A, \quad (1.10)$$

with $B^\# = (B'B)^{-1}B'$ being the left pseudo-inverse of B .

Therefore, rewriting (1.5) as

$$\Pi(s)P(s) = (I - M^{-1}(s))P^\# P,$$

it is readily seen that

$$\Pi = (I - M^{-1})P^\#. \quad (1.11)$$

Referring to the systems depicted in Figure 1.3, the transfer function of the system illustrated in Figure 1.3a is

$$T_{v \rightarrow x}^a = (I - PK)^{-1}P \quad (1.12)$$

And for the second system (Figure 1.3b)

$$T_{v \rightarrow x}^b = (I - PM(K - \Pi))^{-1}PM \quad (1.13)$$

To get the same transfer function as (1.12), one should exploit identity [16]

$$(I + LQ)^{-1}L = L(I + QL)^{-1} \quad (1.14)$$

when L and Q , which may be either singular or nonsingular, satisfy the condition that $I + QL$ is nonsingular, and since the matrices have equal determinants, it follows that $I + LQ$ is also nonsingular. The equation (1.13) may now be rewritten using (1.14) with $L = PM$ and $Q = (K - \Pi)$

$$T_{v \rightarrow x}^b = PM(I - (K - \Pi)PM)^{-1} \quad (1.15)$$

With (1.11)

$$\begin{aligned}
T_{v \rightarrow x}^b &= PM [I - (K - (I - M^{-1})P^\#)PM]^{-1} \\
&= PM [I - KPM + P^\#PM - M^{-1}P^\#PM]^{-1} \\
&= PM [I - KPM + M - I]^{-1} \\
&= PM(-KPM + M)^{-1} \\
&= P(I - KP)^{-1}
\end{aligned}$$

Applying again (1.14) with $L = P$ and $Q = K$ gives the identical closed-loop transfer function as given by (1.12).

This separation can be employed to build controllers, for instance, to mitigate disturbances using a known model via the Internal Model Principle and to counteract unknown disturbances with robust control approaches, such as sliding mode control [19].

Modifying the control scheme may be advantageous for subsequent development. Let now define the state space realisation of the model as

$$M(s) = \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & I \end{array} \right]. \quad (1.16)$$

Integrating (1.3) and (1.11)

$$\begin{aligned}
u &= M(v + (K - (I - M^{-1})P^\#)x) \\
u &= Mv + (MK - MP^\# + MM^{-1}P^\#)x \\
u &= Mv + (K + (M - I)(K - P^\#))x \\
u &= \left[M \quad K + (M - I)(K - P^\#) \right] \begin{bmatrix} v \\ x \end{bmatrix}
\end{aligned} \quad (1.17)$$

Exploitation of formulas for parallel and cascaded systems [38] will be necessary. Begin with the expression

$$(M - I)(K - P^\#) = \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & I - I \end{array} \right] (K - B^\#s + B^\#A). \quad (1.18)$$

Then consider

$$\left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & 0 \end{array} \right] B^\#,$$

and express it as

$$\dot{x}_m = A_m x + B_m B^\# v$$

$$y_m = C_m x_m.$$

Apply the derivative operator s to the output

$$\begin{aligned} z_m = \dot{y}_m &= C_m \dot{x}_m \\ &= C_m A_m x_m + C_m B_m B^\# v \end{aligned}$$

The new system is $(M(s) - I)B^\# s$:

$$(M(s) - I)B^\# s = \left[\begin{array}{c|c} A_m & B_m B^\# \\ \hline C_m A_m & C_m B_m B^\# \end{array} \right],$$

or, equivalently,

$$(M(s) - I)B^\# s = \left[\begin{array}{c|c} A_m & A_m B_m B^\# \\ \hline C_m & C_m B_m B^\# \end{array} \right].$$

Thus, returning to expression (1.18):

$$\begin{aligned} (M - I)(K - P^\#) &= \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & 0 \end{array} \right] (K - B^\# s + B^\# A) \\ &= \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & 0 \end{array} \right] (K + B^\# A) - \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & 0 \end{array} \right] B^\# s \\ &= \left[\begin{array}{c|c} A_m & B_m(K + B^\# A) \\ \hline C_m & 0 \end{array} \right] - \left[\begin{array}{c|c} A_m & A_m B_m B^\# \\ \hline C_m & C_m B_m B^\# \end{array} \right] \\ &= \left[\begin{array}{c|c} A_m & B_m(K + B^\# A) - A_m B_m B^\# \\ \hline C_m & -C_m B_m B^\# \end{array} \right] \end{aligned}$$

and

$$K + (M - I)(K - P^\#) = \left[\begin{array}{c|c} A_m & B_m(K + B^\# A) - A_m B_m B^\# \\ \hline C_m & K - C_m B_m B^\# \end{array} \right].$$

Coming back to (1.17) one gets

$$\begin{aligned}
 & \begin{bmatrix} M & K + (M - I)(K - P^\#) \end{bmatrix} \begin{bmatrix} v \\ x \end{bmatrix} = \\
 & = \left[\left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & I \end{array} \right] \left[\begin{array}{c|c} A_m & B_m(K + B^\# A) - A_m B_m B^\# \\ \hline C_m & K - C_m B_m B^\# \end{array} \right] \right] \begin{bmatrix} v \\ x \end{bmatrix} \\
 & = \left[\begin{array}{c|c|c} A_m & B_m & B_m(K + B^\# A) - A_m B_m B^\# \\ \hline C_m & I & K - C_m B_m B^\# \end{array} \right] \begin{bmatrix} v \\ x \end{bmatrix}.
 \end{aligned}$$

Therefore, the controller is

$$C(s) = \left[\begin{array}{c|c|c} A_m & B_m & B_m(K + B^\# A) - A_m B_m B^\# \\ \hline C_m & I & K - C_m B_m B^\# \end{array} \right]. \quad (1.19)$$

The scheme can be depicted as illustrated in the Figure 1.4.

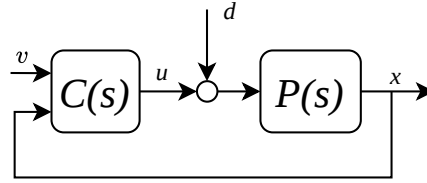


Figure 1.4: New state-feedback scheme

1.3. Illustrative example for state feedback with CIM

Consider the LC-filter scheme shown in Figure 1.5 with parameters in Table 1.1. This kind of filter is commonly used for attenuation of high-frequency distortion in power converters.

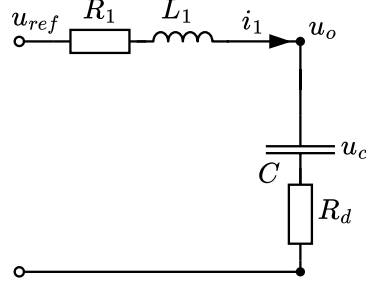


Figure 1.5: Scheme of the LC-filter

Parameter	Value
L_1	$150\mu H$
R_1	0.045Ω
C	$22\mu F$
R_d	1Ω

Table 1.1: Parameters of the LC-filter

Choose the current of the inductor and the voltage of the capacitor as state variables $x = \begin{bmatrix} i_1 & u_c \end{bmatrix}$. The plant can be described by the state equation

$$\dot{x} = Ax + Bu,$$

with

$$A = \begin{bmatrix} -\frac{R_1+R_d}{L_1} & -\frac{1}{L_1} \\ \frac{1}{C} & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} \frac{1}{L_1} \\ 0 \end{bmatrix}.$$

Select $K = \begin{bmatrix} -7.955 & -0.650 \end{bmatrix}$ which places eigenvalues of $A + BK$ to $\{-50000, -10000\}$. The reference signal $u_{ref} = 15 \sin(\omega t)$, where $\omega = 2\pi \text{ rad/s}$. The plant is affected by a disturbance sine wave signal $d = 3 \sin(3\omega t)$ on its input. Here u_{ref} enters through feedforward scaling (u_{req}), so the reference tracking is open-loop; the internal model $M(s)$ is used only for disturbance rejection. Thus, the disturbance can be represented as $D(s) = M(s)D_0(s)$ with

$$M(s) = \frac{(s+a)^2}{s^2 + (3\omega)^2}, \quad d_0(t) = 3(3\omega)^2 t e^{-at},$$

where $a = 8 \cdot 2\pi$. Then

$$\Pi = (I - M^{-1})P^\# = \begin{bmatrix} 0.01508(s + 6967) \\ 100.53 \end{bmatrix} \frac{(s + 21.6)}{(s + 50.27)^2}$$

Applying controller (1.19) as in the scheme in Figure 1.4 with

$$u_{req}(t) = \frac{u_{ref}(t)}{\lim_{s \rightarrow 0} \frac{1}{1 - KP(s)}},$$

the disturbance can be asymptotically rejected, and the reference can be tracked. Simulations were done with MATLAB[®] / Simulink[®]. The results in Figure 1.6 show that the control signal includes the additive inverse of the disturbance, which allows the output voltage to follow the desired reference.

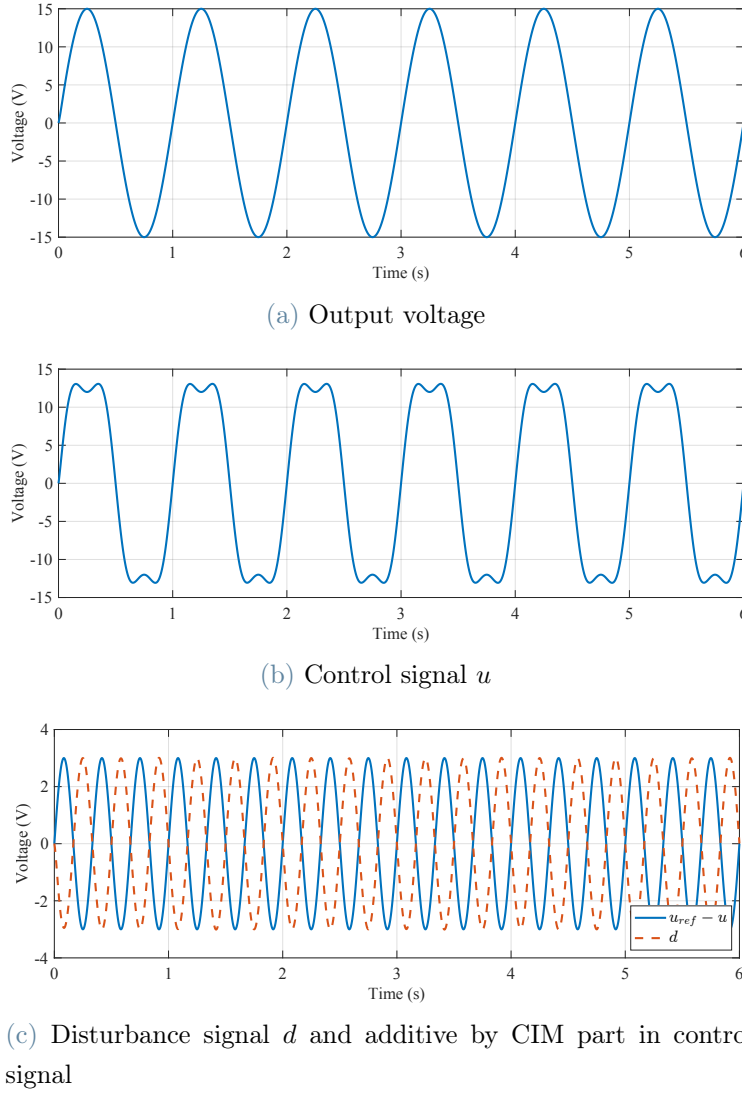


Figure 1.6: Simulation results for CIM with parameter estimation

1.4. IM parameters adaptation for the state feedback

For now, focus on the model parameters in the case of the harmonic external signal. Here, the parameters that should be identified are the frequencies of the signal. Start with a signal that has only one harmonic, and then it may be extended to any number of frequencies. Any harmonic $d = \mu \sin(\omega t + \beta)$ can be represented in the form (1.2) as [19]

$$M(s) = \frac{(s + a)^2}{s^2 + \omega^2} \quad \text{and} \quad d_0(t) = e^{-at}(\alpha t + \beta)\mu, \quad (1.20)$$

where $\alpha = \omega \cos \phi - a \sin \phi$ and $\beta = \sin \phi$. Since $M^{-1} \in H_\infty$ from Assumption 1.1, all zeros of $M(s)$ lie in the open left half-plane; thus $a > 0$. The key trick for the adaptation

is to represent the model affinely in the parameter ω^2 . Consider the model's inverse

$$M^{-1}(s) = \frac{s^2 + \omega^2}{(s+a)^2} = \frac{s^2}{(s+a)^2} + \frac{\omega^2}{(s+a)^2} = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_1 & I \end{array} \right] + \omega^2 \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_2 & 0 \end{array} \right], \quad (1.21)$$

with

$$A_0 = \begin{bmatrix} 0 & 1 \\ -a^2 & -2a \end{bmatrix} \quad B_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C_1 = \begin{bmatrix} -a^2 & -2a \end{bmatrix} \quad C_2 = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

This can be written using different convenient notation.

Notation 1.2. *Lower linear-fractional transformation (LFT) [42] (see Figure 1.7).*

$$\mathcal{LFT}_l \left(\left[\begin{array}{cc} P_{11} & P_{12} \\ P_{21} & P_{22} \end{array} \right], \Theta \right) = P_{11} + P_{12}\Theta(I - P_{22}\Theta)^{-1}P_{21} \quad (1.22)$$

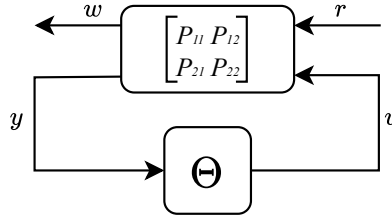


Figure 1.7: Lower linear fractional transformation.

Therefore, $M^{-1}(s)$ can be rewritten as a lower LFT. In view of (1.22), identify

$$\begin{aligned} P_{11} &= \frac{s^2}{(s+a)^2} & P_{12} &= 1 \\ P_{21} &= \frac{1}{(s+a)^2} & P_{22} &= 0 & \Theta &= \omega^2. \end{aligned}$$

Thus,

$$M^{-1}(s) = \mathcal{LFT}_l \left(\left[\begin{array}{c|c} \frac{s^2}{(s+a)^2} & 1 \\ \hline \frac{1}{(s+a)^2} & 0 \end{array} \right], \Theta \right). \quad (1.23)$$

In general, the state space form of the matrix append P from (1.22) looks like [24]

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{cc|cc} A_{11} & & B_{11} & \\ & A_{12} & & B_{12} \\ & & A_{21} & B_{21} \\ & & & A_{22} & B_{22} \\ \hline C_{11} & C_{12} & D_{11} & D_{12} \\ & & C_{21} & C_{22} & D_{21} & D_{22} \end{array} \right]$$

In case of (1.21) with

$$\begin{aligned} P_{11} &= \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_0 & I \end{array} \right] & P_{12} &= \left[\begin{array}{c|c} 0 & 0 \\ \hline 0 & I \end{array} \right] \\ P_{21} &= \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_1 & 0 \end{array} \right] & P_{22} &= \left[\begin{array}{c|c} 0 & 0 \\ \hline 0 & 0 \end{array} \right] \end{aligned}$$

P structure is

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{cc|cc} A_0 & 0 & B_0 & 0 \\ 0 & A_0 & B_0 & 0 \\ \hline C_0 & 0 & I & I \\ 0 & C_1 & 0 & 0 \end{array} \right],$$

which can be simplified to

$$P = \left[\begin{array}{c|cc} A_0 & B_0 & 0 \\ \hline C_0 & I & I \\ C_1 & 0 & 0 \end{array} \right].$$

Therefore,

$$M^{-1}(s) = \mathcal{LFT}_l \left(\left(\left[\begin{array}{c|cc} A_0 & B_0 & 0 \\ \hline C_0 & I & I \\ C_1 & 0 & 0 \end{array} \right], \Theta \right) \right). \quad (1.24)$$

From the inverse model $M^{-1}(s)$, the model $M(s)$ can be recovered as:

$$M(s) = \mathcal{LFT}_l \left(\left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & -B_0 \\ \hline -C_0 & I & -I \\ C_1 & 0 & 0 \end{array} \right], \Theta \right) \right) \quad (1.25)$$

$$M(s) = \left[\begin{array}{c|c} A_0 - B_0(C_0 + C_1\Theta) & B_0 \\ \hline -C_0 - \Theta C_1 & I \end{array} \right] \quad (1.26)$$

The controller can be derived via the same procedure as in (1.17)–(1.19):

$$\begin{aligned}
(M - I)(K - P^\#) &= \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & -B_0 \\ \hline -C_0 & I - I & -I \\ C_1 & 0 & 0 \end{array} \right], \Theta \right) (K - B^\# s + B^\# A) \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & -B_0 \\ \hline -C_0 & 0 & -I \\ C_1 & 0 & 0 \end{array} \right] \left[\begin{array}{cc|c} K - B^\# s + B^\# A & 0 & \\ \hline 0 & & I \end{array} \right], \Theta \right)
\end{aligned} \tag{1.27}$$

For now, consider

$$\left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 \\ \hline -C_0 & 0 \\ C_1 & 0 \end{array} \right] B^\#.$$

Express it as

$$\begin{aligned}
\dot{x}_m &= (A_0 - B_0 C_0)x_m + B_0 B^\# v_m \\
y_m &= \begin{bmatrix} -C_0 \\ C_1 \end{bmatrix} x_m.
\end{aligned}$$

Differentiate the output:

$$\begin{aligned}
z_m = \dot{y}_m &= \begin{bmatrix} -C_0 \\ C_1 \end{bmatrix} \dot{x}_m \\
&= \begin{bmatrix} -C_0 \\ C_1 \end{bmatrix} (A_0 - B_0 C_0)x_m + \begin{bmatrix} -C_0 \\ C_1 \end{bmatrix} B_0 B^\# v_m
\end{aligned}$$

Hence, $(M(s) - I)B^\# s$ admits the realisation:

$$(M(s) - I)B^\# s = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 B^\# \\ \hline -C_0(A_0 - B_0 C_0) & -C_0 B_0 B^\# \\ C_1(A_0 - B_0 C_0) & C_1 B_0 B^\# \end{array} \right]$$

or, equivalently,

$$(M(s) - I)B^\# s = \left[\begin{array}{c|c} A_0 - B_0 C_0 & (A_0 - B_0 C_0)B_0 B^\# \\ \hline -C_0 & -C_0 B_0 B^\# \\ C_1 & C_1 B_0 B^\# \end{array} \right].$$

Thus, returning to expression (1.27),

$$\begin{aligned}
(M - I)(K - P^\#) &= \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & -B_0 \\ \hline -C_0 & 0 & -I \\ C_1 & 0 & 0 \end{array} \right] \left[\begin{array}{cc} K - B^\# s + B^\# A & 0 \\ 0 & I \end{array} \right], \Theta \right) \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & -B_0 \\ \hline -C_0 & 0 & -I \\ C_1 & 0 & 0 \end{array} \right] \left[\begin{array}{cc} K + B^\# A & 0 \\ 0 & I \end{array} \right] \right. \\
&\quad \left. - \left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & -B_0 \\ \hline -C_0 & 0 & -I \\ C_1 & 0 & 0 \end{array} \right] \left[\begin{array}{cc} B^\# s & 0 \\ 0 & 0 \end{array} \right] \right), \Theta \right) \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0(K + B^\# A) & -B_0 \\ \hline -C_0 & 0 & -I \\ C_1 & 0 & 0 \end{array} \right] \right. \\
&\quad \left. - \left[\begin{array}{c|cc} A_0 - B_0 C_0 & (A_0 - B_0 C_0)B_0 B^\# & 0 \\ \hline -C_0 & -C_0 B_0 B^\# & 0 \\ C_1 & C_1 B_0 B^\# & 0 \end{array} \right] \right), \Theta \right) \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0(K + B^\# A) - (A_0 - B_0 C_0)B_0 B^\# & -B_0 \\ \hline -C_0 & C_0 B_0 B^\# & -I \\ C_1 & -C_1 B_0 B^\# & 0 \end{array} \right], \Theta \right).
\end{aligned}$$

Therefore,

$$\begin{aligned}
K + (M - I)(K - P^\#) &= \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0(K + B^\# A) - (A_0 - B_0 C_0)B_0 B^\# & -B_0 \\ \hline -C_0 & K + C_0 B_0 B^\# & -I \\ C_1 & -C_1 B_0 B^\# & 0 \end{array} \right], \Theta \right). \quad (1.28)
\end{aligned}$$

Substituting (1.25) and (1.28) into (1.17) yields

$$\begin{aligned}
\begin{bmatrix} M & K + (M - I)(K - P^\#) \end{bmatrix} \begin{bmatrix} v \\ x \end{bmatrix} &= \\
&= \mathcal{LFT}_l \left(\left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & B_0(K + B^\# A) - (A_0 - B_0 C_0)B_0 B^\# & -B_0 \\ \hline -C_0 & I & K + C_0 B_0 B^\# & -I \\ \hline C_1 & 0 & -C_1 B_0 B^\# & 0 \end{array} \right], \Theta \right) \begin{bmatrix} v \\ x \end{bmatrix}.
\end{aligned}$$

Therefore, the parameter Θ can be time-varying or nonlinear, which allows the introduction of adaptation techniques. The controller $R : (v, x) \mapsto u$ can be defined as $R(s) = \mathcal{LFT}_l(\Psi, \Theta)$, where

$$\Psi(s) = \left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 & B_0(K + B^\# A) - (A_0 - B_0 C_0)B_0 B^\# & -B_0 \\ \hline -C_0 & I & K + C_0 B_0 B^\# & -I \\ \hline C_1 & 0 & -C_1 B_0 B^\# & 0 \end{array} \right]. \quad (1.29)$$

Replacing Θ with its estimation $\hat{\Theta}$ results in the same scheme; moreover, the closed-loop system is stable until $\hat{\Theta}$ is bounded. The scheme can be depicted as illustrated in Figure 1.8.

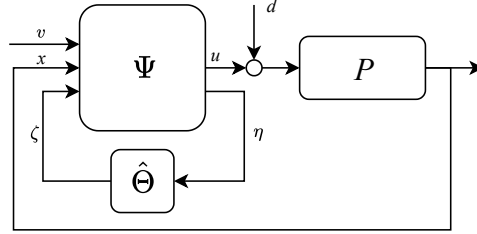


Figure 1.8: Scheme of the adaptive state-feedback controller

1.5. Parameter adaptation

The subsequent result (taken from [8]) defines the stability properties of the adaptive system. Define the tracking error as

$$e(t) := x(t) - x_v(t), \quad (1.30)$$

where the virtual state $x_v(t)$ evolves according to

$$\dot{x}_v(t) = (A + BK)x_v(t) + Bv(t). \quad (1.31)$$

Assume that the matrix $A + BK$ is Hurwitz. Then there exists a symmetric positive definite matrix $P = P' > 0$ that satisfies the Lyapunov inequality

$$(A + BK)'P + P(A + BK) < 0. \quad (1.32)$$

Then, under the adaptation law,

$$\dot{\hat{\Theta}}(t) = B'P e(t) \zeta'(t), \quad \hat{\Theta}(0) = \hat{\Theta}_0, \quad (1.33)$$

where $\zeta(t)$ represents the second output of Ψ shown in Figure 1.8, all signals in the closed-loop system remain globally bounded. Furthermore, the tracking error converges asymptotically to zero; that is,

$$\lim_{t \rightarrow \infty} e(t) = 0.$$

1.6. Illustrative example for parameter adaptation

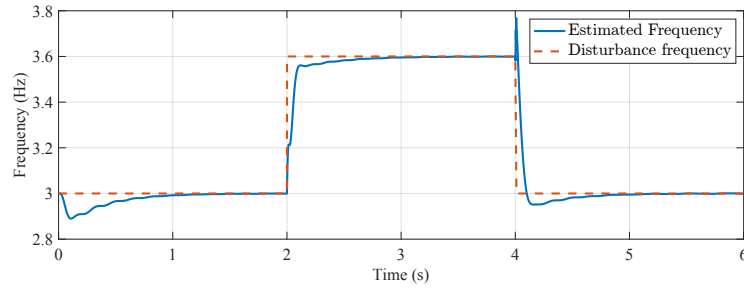
Take the same example as in Section 1.3, but now with a disturbance frequency that switches between $3Hz$ and $3.6Hz$ every 2 seconds. Using the same parameter $a = 8 \cdot 2\pi$ the inverse of the model can be defined as

$$M^{-1}(s) = \mathcal{LFT}_l \left(\left[\begin{array}{c|c} \frac{s^2}{(s+a)^2} & 1 \\ \hline \frac{1}{(s+a)^2} & 0 \end{array} \right], \omega^2 \right) = \mathcal{LFT}_l \left(\left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ -a^2 & -2a & 1 & 0 \\ \hline -a^2 & -2a & 1 & 1 \\ 1 & 0 & 0 & 0 \end{array} \right], \omega^2 \right)$$

Simulations were done with adaptation law (1.33) where $P = \begin{bmatrix} 1283 & 1364 \\ 1364 & 2111 \end{bmatrix}$ is the solution to the Lyapunov equation

$$(A + BK)'P + P(A + BK) < 3 \times 10^7.$$

The parameter converges to the true value as shown in Figure 1.9a. The control signal presented in Figure 1.9c includes the desired reference and the additive inverse of the disturbance $-d(t)$ (Figure 1.9d), allowing the output voltage to be $\lim_{t \rightarrow \infty} u_o(t) = 15 \sin(\omega t)$ (Figure 1.9b).



(a) Disturbance frequency estimation

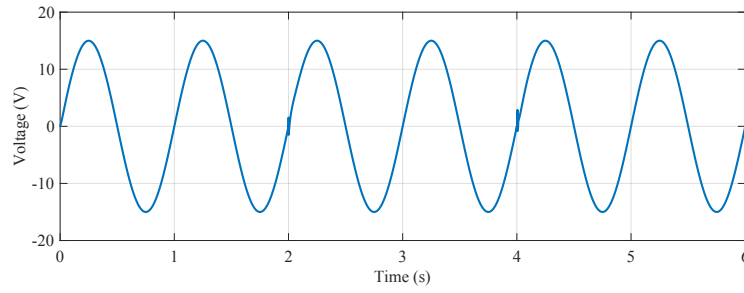
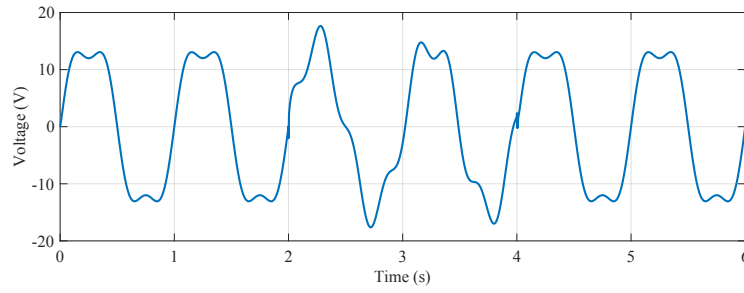
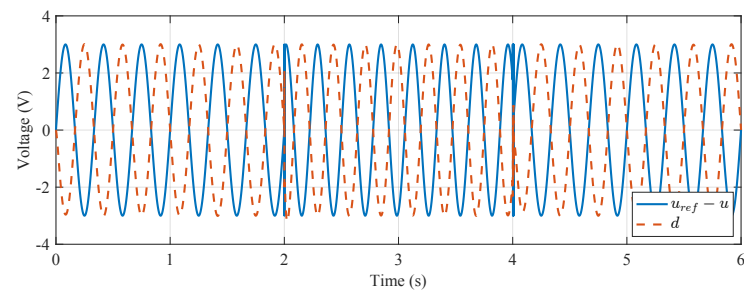
(b) Output voltage u_o (c) Control signal u (d) Disturbance signal d and additive by CIM part in control signal

Figure 1.9: Simulation results for CIM with parameter estimation.

2 | Compensation of internal model for output regulation

Controller design becomes more involved in the case when only a subset of the output signal is regulated [9]. Consider a continuous-time linear time-invariant (LTI) plant P under a control input $u(t) \in \mathbb{R}^m$ and a measured output $y(t) \in \mathbb{R}^p$. The control goal is to reject external disturbances that affect the regulated signal or to track the reference signal. The regulated signal can be defined as

$$y_e(t) = Ey(t) \in \mathbb{R}^{p_e},$$

where $E \in \mathbb{R}^{p_e \times p}$ is assumed to be normalised ($EE' = I$). A possible structure of the controller is [9]

$$R = R_s(E'ME + I - E'E), \quad (2.1)$$

where $M \in \mathbb{R}^{p_e \times p_e}$ is an internal model whose poles coincide with the poles of the external signal. R_s is a stabiliser of the closed loop designed to meet the performance requirements of the system. Each pole of M should have geometric multiplicity p_e for robust regulation [21]. The stabiliser is designed for the augmented plant

$$P_{aug} = (E'ME + I - E'E)P. \quad (2.2)$$

It is always possible to design R_s if there is no unstable cancellation between 2 terms on the right-hand side of (2.2).

Notation 2.1. The closed right half of the complex plane is represented as $\overline{\mathbb{C}}_0$. A' is the complex-conjugate transpose of a matrix A . The notation $\text{spec}(A)$ denotes the matrix spectrum when A is a square matrix or the set of poles when A is an LTI system. The set of bounded and holomorphic functions in the open right-half plane is denoted by H_∞ .

The following assumptions are introduced.

Assumption 2.1.

$$\text{spec}(M) \subset \overline{\mathbb{C}}_0, \quad M^{-1} \in H_\infty, \quad M(\infty) = I.$$

Assumption 2.2.

$$p_e = m \quad \text{and} \quad EP(s) \text{ has full normal rank.}$$

Assumption 2.3.

$$\text{spec}((EP)^{-1}) \cap \text{spec}(M) = \emptyset.$$

The first element of Assumption 2.1 states that stable poles in internal models are unnecessary, as their function is to provide persistent signals of a defined type. The remaining part of that assumption, identical to Assumption 1.1, allows for the integration of controller zeros in $\overline{\mathbb{C}}_0$ or at infinity via R_s , if required for any purpose. Similarly, the normalisation of $M(\infty)$ can consistently be guaranteed by R_s .

Assumption 2.2 asserts that the regulated channel $u \mapsto e$ is neither underactuated, which is essential for any regulatory problem, nor redundant. The latter reduces reasoning; however, it is not strictly required and may be eliminated at the expense of more complex technicalities (see [9]).

Assumption 2.3, in conjunction with the latter part of Assumption 2.1, guarantees the absence of unstable cancellations in (2.2).

2.1. CIM for output regulation

Define the matrix $E_\perp \in \mathbb{R}^{(p-p_e) \times p}$ as any matrix that satisfies

$$E'_\perp E_\perp = I - E'E. \tag{2.3}$$

If $p > p_e$, then every such $E_\perp$ has full row rank and satisfies $E_\perp \begin{bmatrix} E' & E'_\perp \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}$.

The next result can be proved [9].

Theorem 2.1. *R_s internally stabilises P_{aug} defined by (2.2) iff*

$$R_s = \bar{R} (I + E'_\perp \Pi_2 E) - \Pi_1 E, \tag{2.4}$$

for some $\bar{R}$ internally stabilising

$$\bar{P} := (I + E_{\perp} \Pi_2 E) P_{\text{aug}} (I + \Pi_1 E P_{\text{aug}})^{-1}, \quad (2.5)$$

and $\Pi_1, \Pi_2 \in H_{\infty}$ such that $I + \Pi_1 E P_{\text{aug}}$ is invertible.

The introduced internal model compensators Π_1 and Π_2 facilitate the construction of a stabiliser for P_{aug} by enabling the design of a controller for $\bar{P}$. Observe that the configuration of the stabiliser resembles that presented for state-feedback control (1.4), with the distinction that, in addition to the parallel compensator Π_1 , a cascaded term with Π_2 is incorporated. It is easily apparent that when $y_e = y$, the $E_{\perp}$ matrix is null, and the stabiliser has an identical structure to (1.4).

Furthermore, a key result is that $\bar{P}$ can always be chosen to be of the same order as P . This follows from the following conclusions. $\bar{P}$ should be such that [9]

$$\begin{bmatrix} \Pi_1 \\ \Pi_2 \end{bmatrix} = \begin{bmatrix} I \\ E_{\perp} \bar{P} \end{bmatrix} (E \bar{P})^{-1} - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M^{-1} \quad (2.6)$$

are stable. In addition, $I + \Pi_1 M E P$ should be invertible, and it is always true if $EP(s)$ is strictly proper.

$(E \bar{P})^{-1}$ and $E_{\perp} \bar{P} (E \bar{P})^{-1}$ should match unstable, including non-proper parts of $(EP)^{-1} M^{-1}$ and $E_{\perp} P (EP)^{-1} M^{-1}$, respectively. Nevertheless, given that M^{-1} is stable according to Assumption 2.1, it follows that only the plant P contributes to instabilities. Consequently, the complexity of $\bar{P}$ must not exceed that of P . If EP is stably invertible with relative degree 1 (harmonic internal model), then the natural selection is $\bar{P} = P$. Conversely, the relationship between $\bar{P}$ and P is typically more complex.

Regarding closed-loop stability. The closed-loop sensitivity, S , and disturbance sensitivity T_d are given by

$$\begin{bmatrix} S & T_d \end{bmatrix} = (I - PR)^{-1} \begin{bmatrix} I & P \end{bmatrix}.$$

It follows from [9] that

$$E \begin{bmatrix} S & T_d \end{bmatrix} = M^{-1} E \begin{bmatrix} \bar{S} & \bar{T}_d \end{bmatrix} \begin{bmatrix} I - P(EP)^{-1} E & 0 \\ (EP)^{-1} E & I \end{bmatrix}, \quad (2.7)$$

meaning that unstable poles of $M(s)$ are zeros of both $ES(s)$ and $ET_d(s)$, which enable the asymptotic rejection of persistent disturbances and/or tracking signals.

Define state-space realisations of the plant P and the internal model M by analogy with (1.9) and (1.16),

$$P(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right], \quad M(s) = \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & I \end{array} \right], \quad (2.8)$$

with state dimensions n and n_m , respectively. Assume

Assumption 2.4.

$$ED = 0,$$

which means that $P_e(s) = EP(s)$ is strictly proper.

Assumption 2.5.

Zeros of $M(s)$ and $P_e(s)$ are disjoint.

Selecting (stable) zeros of $M(s)$ that differ from those of $P_e(s)$ is not restrictive; this can be compensated by R_s . Also introduce matrices $B^\# \in \mathbb{R}^{m \times n}$ and $B^\perp \in \mathbb{R}^{(n-m) \times n}$ (a matrix whose rows are orthogonal to the columns of B) such that

$$\begin{bmatrix} B^\perp \\ B^\# \end{bmatrix} B = \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad \det \begin{bmatrix} B^\perp \\ B^\# \end{bmatrix} \neq 0. \quad (2.9)$$

They exist whenever B has full column rank, which is guaranteed by Assumptions 2.2 and 2.4. The following result can then be proved [9].

Proposition 2.1. *Under Assumptions 2.1–2.5 the generalised Sylvester equation*

$$\begin{bmatrix} B^\perp \\ 0 \end{bmatrix} X(A_m - B_m C_m) - \begin{bmatrix} B^\perp A \\ -EC \end{bmatrix} X = \begin{bmatrix} 0 \\ C_m \end{bmatrix} \quad (2.10)$$

has a unique bounded solution $X \in \mathbb{R}^{n \times n_m}$ and

$$\begin{bmatrix} \Pi_1(s) \\ \Pi_2(s) \end{bmatrix} = \left[\begin{array}{c|c} \frac{A_m - B_m C_m}{C_\Pi} & B_m \\ \hline E_\perp C X + E_\perp D C_\Pi & 0 \end{array} \right], \quad (2.11)$$

where

$$C_\Pi := B^\# X(A_m - B_m C_m) - B^\# A X, \quad (2.12)$$

and

$$\bar{P}(s) = \left[\begin{array}{c|c} \frac{A + XB_mEC}{C} & \frac{B}{D} \end{array} \right] \quad (2.13)$$

satisfies (2.6).

Using the expressions from Theorem 2.1 and Proposition 2.1, derive the state-space realisation of the controller (2.1). Combining (2.1) and (2.4) yields

$$\begin{aligned} R &= (\bar{R}(I + E'_\perp \Pi_2 E) - \Pi_1 E)(E' ME + I - E' E) \\ &= \bar{R}(I + E'_\perp \Pi_2 E)(E' ME + I - E' E) - \Pi_1 E(E' ME + I - E' E) \\ &= \bar{R}(E' ME + I - E' E + E'_\perp \Pi_2 EE' ME + E'_\perp \Pi_2 E - E'_\perp \Pi_2 EE' E) \\ &\quad - \Pi_1 (EE' ME + EI - EE' E). \end{aligned}$$

Since E is defined so that $EE' = I$, the expression can be simplified to

$$R = \bar{R}(E' ME + I - E' E + E'_\perp \Pi_2 ME) - \Pi_1 ME. \quad (2.14)$$

A state-space realisation is obtained using standard rules for series and parallel interconnections [38]. Consider first the term $\Pi_1 ME$. Using (2.8) and the first output of (2.11), it follows that

$$\begin{aligned} \Pi_1 ME &= \left[\begin{array}{c|c} \frac{A_m - B_m C_m}{C_\Pi} & \frac{B_m}{0} \end{array} \right] \left[\begin{array}{c|c} \frac{A_m}{C_m} & \frac{B_m}{I} \end{array} \right] E \\ &= \left[\begin{array}{c|c} \frac{A_m}{B_m C_m} & \frac{0}{A_m - B_m C_m} \\ \hline \frac{0 C_m}{C_\Pi} & \frac{B_m E}{0 E} \end{array} \right], \end{aligned}$$

which, after a basic similarity transformation, reduces to

$$\Pi_1 ME = \left[\begin{array}{c|c} \frac{A_m}{C_\Pi} & \frac{B_m}{I} \end{array} \right]. \quad (2.15)$$

Next, for the first factor in the stabiliser term, one obtains

$$\begin{aligned} E' ME + I - E' E &= \left[\begin{array}{c|c} \frac{A_m}{EC_m} & \frac{B_m E}{E' E + I - E' E} \end{array} \right] \\ &= \left[\begin{array}{c|c} \frac{A_m}{E' C_m} & \frac{B_m E}{I} \end{array} \right]. \end{aligned}$$

The latter one with (2.8) and the second output of (2.11) is

$$E'_\perp \Pi_2 M E = \left[\begin{array}{c|c} \frac{A_m}{B_m C_m} & 0 \\ \hline \frac{B_m C_m}{E'_\perp 0 C_m} & \frac{A_m - B_m C_m}{E'_\perp (E_\perp C X + E_\perp D C_\Pi)} \end{array} \middle| \begin{array}{c} B_m E \\ B_m E \\ 0 E \end{array} \right],$$

which, again, can be simplified to

$$E'_\perp \Pi_2 M E = \left[\begin{array}{c|c} A_m & B_m E \\ \hline E'_\perp E'_\perp C X + E'_\perp E_\perp D C_\Pi & 0 \end{array} \right]. \quad (2.16)$$

Therefore,

$$E' M E + I - E' E + E'_\perp \Pi_2 M E = \left[\begin{array}{c|c} A_m & B_m E \\ \hline E' C_m + E'_\perp E_\perp C X + E'_\perp E_\perp D C_\Pi & I \end{array} \right].$$

From the second row of the Sylvester equation (2.10) $C_m = E C X$. Moreover, $E' E = I - E'_\perp E_\perp$ (see (2.3)). Thus, the term in the output can be simplified to

$$\begin{aligned} E' C_m + E'_\perp E_\perp C X + E'_\perp E_\perp D C_\Pi &= (I - E'_\perp E_\perp) C X + E'_\perp E_\perp C X + E'_\perp E_\perp D C_\Pi \\ &= C X + E'_\perp E_\perp D C_\Pi \end{aligned}$$

and again,

$$E' M E + I - E' E + E'_\perp \Pi_2 M E = \left[\begin{array}{c|c} A_m & B_m E \\ \hline C X + E'_\perp E_\perp D C_\Pi & I \end{array} \right]. \quad (2.17)$$

Substituting (2.15) and (2.17) into (2.14) yields

$$R(s) = \begin{bmatrix} \bar{R}(s) & I \end{bmatrix} \left[\begin{array}{c|c} A_m & B_m E \\ \hline C X + E'_\perp E_\perp D C_\Pi & I \\ -C_\Pi & 0 \end{array} \right]. \quad (2.18)$$

The resulting two-degree-of-freedom (2DOF) architecture is shown in Figure 2.1.

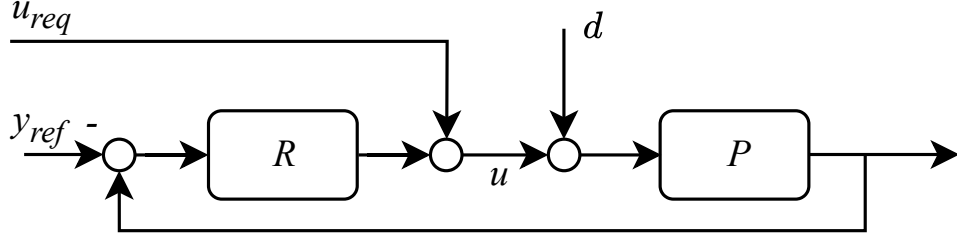


Figure 2.1: Scheme of the 2DOF controller with CIM.

2.2. Parameters adaptation for output regulation

To enable adaptation, the same lower LFT parametrisation as in Section 1.4 is used, but with the parameter appearing on the right (see [20]):

$$M^{-1}(s) = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_0 & I \end{array} \right] + \left[\begin{array}{c|c} A_0 & B_1 \\ \hline C_0 & 0 \end{array} \right] \Theta = M_0(s) + M_1(s)\Theta, \quad (2.19)$$

for some $M_0, M_1 \in H_\infty$ and the static unknown parameter $\Theta \in \mathbb{R}^{v \times m}$, with a number of uncertain parameters $v \in \mathbb{N}$. Therefore, equation (2.6) yields

$$\begin{bmatrix} \Pi_1 \\ \Pi_2 \end{bmatrix} = \begin{bmatrix} I \\ E_\perp \bar{P} \end{bmatrix} (E\bar{P})^{-1} - \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} M_0 - \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} M_1 \Theta, \quad (2.20)$$

However, for an equivalent plant to be independent of Θ , the following sufficient assumption is introduced [20]:

Assumption 2.6.

$EP(s)$ is minimum-phase with a relative degree of at most 2, and $E_\perp P(s)$ has no unstable poles that are not poles of $EP(s)$.

This assumption guarantees that $(EP)^{-1}M_1$ and $E_\perp P(EP)^{-1}M_1$ are stable, and Theorem 2.1 can be stated as follows:

Theorem 2.2. *Under Assumption 2.6 $\tilde{R}_s$, internally stabilises P_{aug} defined by (2.2) if*

$$\tilde{R}_s = \tilde{R}(I + E'_\perp \tilde{\Pi}_2 E) - \tilde{\Pi}_1 E, \quad (2.21)$$

for some $\tilde{R}$ internally stabilising

$$\tilde{P} := (I + E_{\perp} \hat{\Pi}_2 E) \hat{P}_{\text{aug}} (I + \hat{\Pi}_1 E \hat{P}_{\text{aug}})^{-1}, \quad (2.22)$$

where

$$\hat{P}_{\text{aug}} := (E' M_0 E + I - E' E) P, \quad (2.23)$$

and $\hat{\Pi}_1, \hat{\Pi}_2 \in H_{\infty}$ are such that $I + \hat{\Pi}_1 E \hat{P}_{\text{aug}}$ is invertible and

$$\begin{bmatrix} \tilde{\Pi}_1 \\ \tilde{\Pi}_2 \end{bmatrix} = \begin{bmatrix} \hat{\Pi}_1 \\ \hat{\Pi}_2 \end{bmatrix} - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M_1 \Theta. \quad (2.24)$$

Therefore, the controller of the original plant (2.1) is defined as

$$R = \tilde{R}_s (E' M E + I - E' E). \quad (2.25)$$

The model is written as

$$M^{-1}(s) = M_0(s) + M_1(s) \hat{\Theta}, \quad (2.26)$$

where $\hat{\Theta} \in \mathbb{R}^{v \times m}$ is the estimate of the unknown parameter Θ . An equivalent plant $\tilde{P}$ is constructed so that in

$$\begin{bmatrix} \hat{\Pi}_1 \\ \hat{\Pi}_2 \end{bmatrix} = \begin{bmatrix} I \\ E_{\perp} \tilde{P} \end{bmatrix} (E \tilde{P})^{-1} - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M_0 \in H_{\infty} \quad (2.27)$$

$\tilde{P}$ cancels all instabilities in the terms $(EP)^{-1} M_0$ and $E_{\perp} P (EP)^{-1} M_0$.

Moreover, Proposition 2.1 can be rewritten independently of Θ .

Proposition 2.2. *Under Assumptions 2.1–2.6, the generalised Sylvester equation*

$$\begin{bmatrix} B^{\perp} \\ 0 \end{bmatrix} X A_0 - \begin{bmatrix} B^{\perp} A \\ -EC \end{bmatrix} X = \begin{bmatrix} 0 \\ C_0 \end{bmatrix} \quad (2.28)$$

has a unique bounded solution $X \in \mathbb{R}^{n \times n_m}$ and

$$\begin{bmatrix} \hat{\Pi}_1(s) \\ \hat{\Pi}_2(s) \end{bmatrix} = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline \hat{C}_{\Pi} & 0 \\ E_{\perp} C X + E_{\perp} D \hat{C}_{\Pi} & 0 \end{array} \right], \quad (2.29)$$

where

$$\hat{C}_\Pi := B^\# X A_0 - B^\# A X, \quad (2.30)$$

and

$$\tilde{P}(s) = \left[\begin{array}{c|c} \frac{A + X B_0 E C}{C} & \frac{B}{D} \end{array} \right] \quad (2.31)$$

satisfies (2.27).

In order to obtain the controller expression, substitute $\bar{P}$ in (2.20) with $\tilde{P}$

$$\begin{bmatrix} \tilde{\Pi}_1 \\ \tilde{\Pi}_2 \end{bmatrix} = \begin{bmatrix} I \\ E_\perp \tilde{P} \end{bmatrix} (E \tilde{P})^{-1} - \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} M_0 - \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} M_1 \hat{\Theta}. \quad (2.32)$$

In this case, the controller for the original plant is obtained from (2.25) using (2.21) and (2.32):

$$\begin{aligned} R &= \tilde{R}(E' M E + I - E' E + E'_\perp \tilde{\Pi}_2 M E) - \tilde{\Pi}_1 M E \\ &= \tilde{R} - \tilde{R} E' E + [\tilde{R}(E' + E'_\perp \tilde{\Pi}_2) - \tilde{\Pi}_1] M E \\ &= \tilde{R} - \tilde{R} E' E + \left[\tilde{R} \left[E' + E'_\perp \left(E_\perp \tilde{P} (E \tilde{P})^{-1} - E_\perp P (EP)^{-1} M_0 - E_\perp P (EP)^{-1} M_1 \hat{\Theta} \right) \right] \right. \\ &\quad \left. - \left((E \tilde{P})^{-1} - (EP)^{-1} M_0 - (EP)^{-1} M_1 \hat{\Theta} \right) \right] M E \end{aligned}$$

Under (2.3), and collecting terms with $\hat{\Theta}$, one obtains:

$$\begin{aligned} R &= \tilde{R} - \tilde{R} E' E + \left[\tilde{R} \left[E' + (I - E' E) \left(\tilde{P} (E \tilde{P})^{-1} - P (EP)^{-1} M_0 \right) \right] \right. \\ &\quad \left. - (E \tilde{P})^{-1} + (EP)^{-1} M_0 + \left((EP)^{-1} - \tilde{R} (I - E' E) P (EP)^{-1} \right) M_1 \hat{\Theta} \right] (M_0 + M_1 \hat{\Theta})^{-1} E \\ &= \tilde{R} - \tilde{R} E' E + \left[\tilde{R} \left[\tilde{P} (E \tilde{P})^{-1} - P (EP)^{-1} M_0 + E' M_0 \right] \right. \\ &\quad \left. - (E \tilde{P})^{-1} + (EP)^{-1} M_0 + \left((EP)^{-1} - \tilde{R} P (EP)^{-1} + \tilde{R} E' \right) M_1 \hat{\Theta} \right] (M_0 + M_1 \hat{\Theta})^{-1} E \end{aligned}$$

Consider now the following Lemma [42].

Lemma 2.1. *Suppose C is invertible. Then*

$$\begin{aligned} (A + BQ)(C + DQ)^{-1} &= \mathcal{LFT}_\ell(M, Q), \\ (C + QD)^{-1}(A + QB) &= \mathcal{LFT}_\ell(N, Q), \end{aligned}$$

with

$$M = \begin{bmatrix} AC^{-1} & B - AC^{-1}D \\ C^{-1} & -C^{-1}D \end{bmatrix}, \quad N = \begin{bmatrix} C^{-1}A & C^{-1} \\ B - DC^{-1}A & -DC^{-1} \end{bmatrix}.$$

Since M_0 is invertible, the second term can be transformed into lower LFT defining

$$\begin{aligned} A &= \tilde{R} \left[\tilde{P}(E\tilde{P})^{-1} - P(EP)^{-1}M_0 + E'M_0 \right] - (E\tilde{P})^{-1} + (EP)^{-1}M_0, \\ B &= \left((EP)^{-1} - \tilde{R}P(EP)^{-1} + \tilde{R}E' \right) M_1, \\ C &= M_0, \\ D &= M_1. \end{aligned}$$

The elements in the matrix are

$$\begin{aligned} M_{11} &= AC^{-1} = \left[\tilde{R} \left[\tilde{P}(E\tilde{P})^{-1} - P(EP)^{-1}M_0 + E'M_0 \right] - (E\tilde{P})^{-1} + (EP)^{-1}M_0 \right] M_0^{-1} \\ &= \tilde{R}\tilde{P}(E\tilde{P})^{-1}M_0^{-1} - \tilde{R}P(EP)^{-1} + \tilde{R}E' - (E\tilde{P})^{-1}M_0^{-1} + (EP)^{-1} \\ &= (I - \tilde{R}P)(EP)^{-1} - (I - \tilde{R}\tilde{P})(E\tilde{P})^{-1}M_0^{-1} + \tilde{R}E' \\ &= (I - \tilde{R}P)(EP)^{-1} - (E\bar{T}_d)^{-1}M_0^{-1} + \tilde{R}E', \\ M_{12} &= B - AC^{-1}D = \left((EP)^{-1} - \tilde{R}P(EP)^{-1} + \tilde{R}E' \right) M_1 \\ &\quad - \left(\tilde{R}\tilde{P}(E\tilde{P})^{-1}M_0^{-1} - \tilde{R}P(EP)^{-1} + \tilde{R}E' - (E\tilde{P})^{-1}M_0^{-1} + (EP)^{-1} \right) M_1 \\ &= (I - \tilde{R}\tilde{P})(E\tilde{P})^{-1}M_0^{-1}M_1, \\ M_{21} &= M_0^{-1}, \\ M_{22} &= -M_0^{-1}M_1, \end{aligned}$$

where $E\bar{T}_d = (I - \tilde{P}\tilde{R})^{-1}EP$ is the output sensitivity function of the regulated signal.

Therefore,

$$\begin{aligned} R &= \tilde{R} - \tilde{R}E'E + \mathcal{LFT}_l \left(\begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}, \hat{\Theta} \right) E \\ &= \tilde{R} - \tilde{R}E'E + \mathcal{LFT}_l \left(\begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} E & 0 \\ 0 & I \end{bmatrix}, \hat{\Theta} \right) \\ &= \tilde{R} - \tilde{R}E'E + \mathcal{LFT}_l \left(\begin{bmatrix} M_{11}E & M_{12} \\ M_{21}E & M_{22} \end{bmatrix}, \hat{\Theta} \right) \\ &= \mathcal{LFT}_l \left(\begin{bmatrix} \tilde{R} - \tilde{R}E'E + M_{11}E & M_{12} \\ M_{21}E & M_{22} \end{bmatrix}, \hat{\Theta} \right). \end{aligned}$$

Terms in the left column are

$$\begin{aligned} M_{11} &= \tilde{R} - \tilde{R}E'E + (I - \tilde{R}P)(EP)^{-1}E - (E\bar{T}_d)^{-1}M_0^{-1}E + \tilde{R}E'E \\ &= \tilde{R} + (I - \tilde{R}P)(EP)^{-1}E - (E\bar{T}_d)^{-1}M_0^{-1}E, \\ M_{21} &= M_0^{-1}E. \end{aligned}$$

Finally, the controller is defined as

$$R = \mathcal{LFT}_l(\Psi, \hat{\Theta}), \quad (2.33)$$

where

$$\Psi = \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix}, \quad (2.34)$$

with

$$\begin{aligned} \Psi_{11} &= \tilde{R} + (I - \tilde{R}P)(EP)^{-1}E - (E\bar{T}_d)^{-1}M_0^{-1}E, \\ \Psi_{12} &= (I - \tilde{R}\tilde{P})(E\tilde{P})^{-1}M_0^{-1}M_1, \\ \Psi_{21} &= M_0^{-1}E, \\ \Psi_{22} &= -M_0^{-1}M_1. \end{aligned} \quad (2.35)$$

The controller scheme is shown in Figure 2.2.

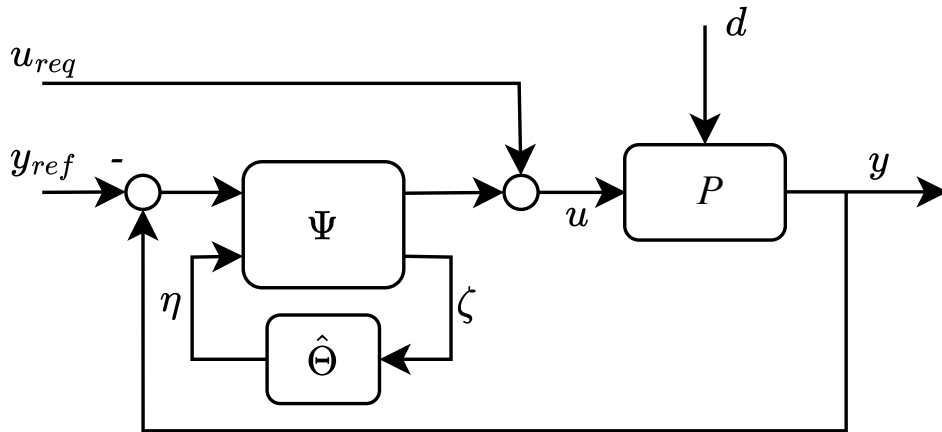


Figure 2.2: Scheme of the 2DOF controller with CIM and affine model parameters

Separate the regulator from the internal model and compensators in the definition of Ψ

definition (2.34)–(2.35). The term Ψ_{11} can be rewritten as:

$$\begin{aligned}\Psi_{11} &= \tilde{R} + (EP)^{-1}E - \tilde{R}P(EP)^{-1}E - (E\tilde{P})^{-1}M_0^{-1}E + \tilde{R}\tilde{P}(E\tilde{P})^{-1}M_0^{-1}E \\ &= \begin{bmatrix} I & \tilde{R} \end{bmatrix} \begin{bmatrix} (EP)^{-1}E - (E\tilde{P})^{-1}M_0^{-1}E \\ I - P(EP)^{-1}E + \tilde{P}(E\tilde{P})^{-1}M_0^{-1}E \end{bmatrix}\end{aligned}$$

Rewrite the second matrix using the $\hat{\Pi}_1$ and $\hat{\Pi}_2$ definitions (2.27). The first row of the second term can be rearranged to

$$\begin{aligned}(EP)^{-1}E - (E\tilde{P})^{-1}M_0^{-1}E &= ((EP)^{-1}M_0E - (E\tilde{P})^{-1})M_0^{-1}E \\ &= -\hat{\Pi}_1M_0^{-1}E.\end{aligned}$$

The second row modification is more involved. Multiply it by $E'E + E'_\perp E_\perp = I$ (see (2.3)) on the left

$$\begin{aligned}(E'E + E'_\perp E_\perp)(I - P(EP)^{-1}E + \tilde{P}(E\tilde{P})^{-1}M_0^{-1}E) \\ &= E'E + E'_\perp E_\perp - E'EP(EP)^{-1}E - E'_\perp E_\perp P(EP)^{-1}E \\ &\quad + E'E\tilde{P}(E\tilde{P})^{-1}M_0^{-1}E + E'_\perp E_\perp \tilde{P}(E\tilde{P})^{-1}M_0^{-1}E \\ &= E'E + E'_\perp E_\perp - E'E - E'_\perp E_\perp P(EP)^{-1}E \\ &\quad + E'M_0^{-1}E + E'_\perp E_\perp \tilde{P}(E\tilde{P})^{-1}M_0^{-1}E \\ &= \left(E' + E_\perp(E_\perp \tilde{P}(E\tilde{P})^{-1} - E_\perp P(EP)^{-1})\right)M_0^{-1}E + E'_\perp E_\perp \\ &= \left(E' + E_\perp \hat{\Pi}_2\right)M_0^{-1}E + E'_\perp E_\perp.\end{aligned}$$

Thus,

$$\Psi_{11} = \begin{bmatrix} I & \tilde{R} \end{bmatrix} \begin{bmatrix} -\hat{\Pi}_1 \\ E' + E_\perp \hat{\Pi}_2 \end{bmatrix} M_0^{-1}E + \begin{bmatrix} 0 \\ E'_\perp E_\perp \end{bmatrix}.$$

Other elements of Ψ are

$$\begin{aligned}\Psi_{12} &= ((E\tilde{P})^{-1} - \tilde{R}\tilde{P}(E\tilde{P})^{-1})M_0^{-1}M_1 \\ &= \begin{bmatrix} I & \tilde{R} \end{bmatrix} \begin{bmatrix} (E\tilde{P})^{-1} \\ -\tilde{P}(E\tilde{P})^{-1} \end{bmatrix} M_0^{-1}M_1, \\ \Psi_{21} &= I(M_0^{-1}E), \\ \Psi_{22} &= -I(M_0^{-1}M_1).\end{aligned}$$

Finally, the resulting structure (see Figure 2.3) is

$$\Psi = \begin{bmatrix} I & \tilde{R} & 0 \\ 0 & 0 & I \end{bmatrix} \Psi_0, \quad (2.36)$$

where

$$\Psi_0 = \begin{bmatrix} -\hat{\Pi}_1 M_0^{-1} E & (E\tilde{P})^{-1} M_0^{-1} M_1 \\ (E' + E_\perp \hat{\Pi}_2) M_0^{-1} E + E'_\perp E_\perp & -\tilde{P}(E\tilde{P})^{-1} M_0^{-1} M_1 \\ M_0^{-1} E & -M_0^{-1} M_1 \end{bmatrix}, \quad (2.37)$$

or, equivalently, in a more structured form:

$$\Psi_0 = \begin{bmatrix} -\hat{\Pi}_1 & (E\tilde{P})^{-1} \\ E' + E_\perp \hat{\Pi}_2 & -\tilde{P}(E\tilde{P})^{-1} \\ I & -I \end{bmatrix} \begin{bmatrix} M_0^{-1} E & 0 \\ 0 & M_0^{-1} M_1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ E'_\perp E_\perp & 0 \\ 0 & 0 \end{bmatrix} \quad (2.38)$$

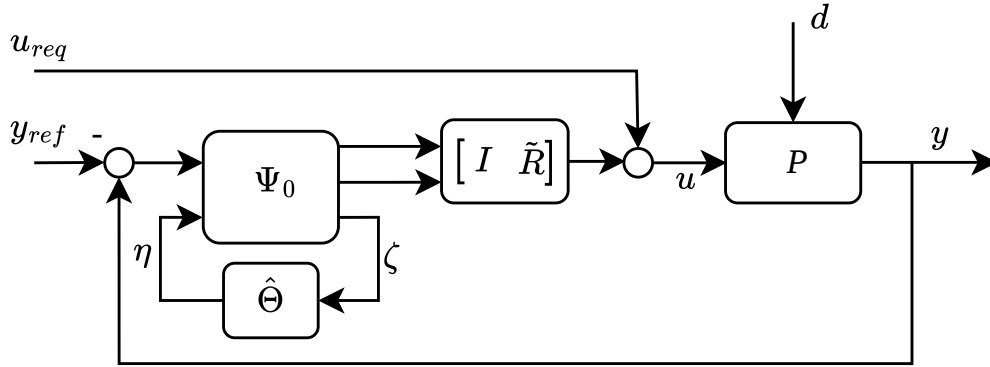


Figure 2.3: Scheme of the 2DOF controller with CIM and separated stabiliser.

State space realisation of the controller is derived based on the form (2.37) with elements

$$\begin{aligned} \Psi_{011} &= \begin{bmatrix} -\hat{\Pi}_1 M_0^{-1} E \\ (E' + E_\perp \hat{\Pi}_2) M_0^{-1} E + E'_\perp E_\perp \end{bmatrix}, \\ \Psi_{012} &= \begin{bmatrix} (E\tilde{P})^{-1} M_0^{-1} M_1 \\ -\tilde{P}(E\tilde{P})^{-1} M_0^{-1} M_1 \end{bmatrix}, \\ \Psi_{021} &= M_0^{-1} E, \\ \Psi_{022} &= -M_0^{-1} M_1. \end{aligned} \quad (2.39)$$

Due to the fact that EP is strictly proper (Assumption 2.4), its inverse does not have a standard state space representation. Thus, introduce a notation for improper systems.

Notation 2.2. *The descriptor state space can be used to represent an improper system by state-space method [33]:*

$$\begin{aligned} E\dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

Rewrite the state space as

$$\begin{aligned} E\dot{x} &= Ax + Bu + 0y \\ 0\dot{u} &= -Cx - Du + y. \end{aligned}$$

Then its inverse is defined as [24]

$$\begin{aligned} \underbrace{\begin{bmatrix} E & 0 \\ 0 & 0 \end{bmatrix}}_{\text{New } E} \underbrace{\begin{bmatrix} \dot{x} \\ \dot{u} \end{bmatrix}}_{\text{New } A} &= \underbrace{\begin{bmatrix} A & B \\ -C & -D \end{bmatrix}}_{\text{New } A} \underbrace{\begin{bmatrix} x \\ u \end{bmatrix}}_{\text{New } B} + \underbrace{\begin{bmatrix} 0 \\ I \end{bmatrix}}_{\text{New } B} y, \\ u &= \underbrace{\begin{bmatrix} 0 & I \end{bmatrix}}_{\text{New } C} \underbrace{\begin{bmatrix} x \\ u \end{bmatrix}}_{\text{New } D} + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_{\text{New } D} y. \end{aligned} \tag{2.40}$$

The compact notation is used:

$$\left[\begin{array}{c|c|c} E & A & B \\ \hline & C & D \end{array} \right] = D + C(sE - A)^{-1}B$$

for transfer functions in terms of their state-space realisations. Hereafter $E = I$ for proper systems.

Plant itself and equivalent plant are defined as (see (2.8), (2.31))

$$P(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \quad \text{and} \quad \tilde{P} = \left[\begin{array}{c|c} A + XB_0EC & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|c} \tilde{A} & B \\ \hline C & D \end{array} \right].$$

According to (2.29), internal model compensators independent of $\hat{\Theta}$ are

$$\hat{\Pi}_1(s) = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline \hat{C}_\Pi & 0 \end{array} \right] \quad \text{and} \quad \hat{\Pi}_2(s) = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline E_\perp CX + E_\perp D\hat{C}_\Pi & 0 \end{array} \right] = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline \hat{C}_{\Pi_2} & 0 \end{array} \right].$$

Systems M_0 and M_1 are given by (2.19)

$$M_0 = \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_0 & I \end{array} \right] \quad \text{and} \quad M_1 = \left[\begin{array}{c|c} A_0 & B_1 \\ \hline C_0 & 0 \end{array} \right].$$

Start by taking the inverse of M_0

$$M_0^{-1} = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 \\ \hline -C_0 & I \end{array} \right].$$

Then Ψ_{021} is

$$\Psi_{021} = M_0^{-1} E = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline -C_0 & E \end{array} \right].$$

Multiply M_0^{-1} and M_1

$$M_0^{-1} M_1 = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 \\ \hline -C_0 & I \end{array} \right] \left[\begin{array}{c|c} A_0 & B_1 \\ \hline C_0 & 0 \end{array} \right] = \left[\begin{array}{cc|c} A_0 - B_0 C_0 & B_0 C_0 & 0 \\ 0 & A_0 & B_1 \\ \hline -C_0 & I C_0 & 0 \end{array} \right]. \quad (2.41)$$

Using similarity transformation $T = \begin{bmatrix} I & -I \\ 0 & I \end{bmatrix}$,

$$M_0^{-1} M_1 = \left[\begin{array}{c|c} A_0 - B_0 C_0 & -B_1 \\ \hline -C_0 & 0 \end{array} \right]. \quad (2.42)$$

Then Ψ_{022} is

$$\Psi_{022} = -M_0^{-1} M_1 = \left[\begin{array}{c|c} A_0 - B_0 C_0 & -B_1 \\ \hline C_0 & 0 \end{array} \right].$$

Append matrices Ψ_{021} and Ψ_{022} :

$$\begin{aligned} \begin{bmatrix} \Psi_{021} & \Psi_{022} \end{bmatrix} &= \left[\begin{array}{cc|cc} A_0 - B_0 C_0 & 0 & B_0 E & 0 \\ 0 & A_0 - B_0 C_0 & 0 & -B_1 \\ \hline -C_0 & C_0 & E & 0 \end{array} \right] \\ &= \left[\begin{array}{c|cc} A_0 - B_0 C_0 & B_0 E & B_1 \\ \hline -C_0 & E & 0 \end{array} \right] \end{aligned}$$

Now focus on the term Ψ_{011} :

$$\begin{aligned} -\hat{\Pi}_1 M_0^{-1} E &= \left[\begin{array}{c|c} A_0 & B_0 \\ \hline -\hat{C}_\Pi & 0 \end{array} \right] \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline -C_0 & E \end{array} \right] \\ &= \left[\begin{array}{cc|c} A_0 & B_0(-C_0) & B_0 E \\ 0 & A_0 - B_0 C_0 & B_0 E \\ \hline -\hat{C}_\Pi & 0 & 0 \end{array} \right]. \end{aligned}$$

Now apply the similarity transformation:

$$-\hat{\Pi}_1 M_0^{-1} E = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline -\hat{C}_\Pi & 0 \end{array} \right]$$

Meanwhile, the second row is

$$\begin{aligned} (E' + E_\perp \hat{\Pi}_2) M_0^{-1} E + E'_\perp E_\perp &= \left(E' + E_\perp \left[\begin{array}{c|c} A_0 & B_0 \\ \hline \hat{C}_{\Pi_2} & 0 \end{array} \right] \right) \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline -C_0 & E \end{array} \right] + E'_\perp E_\perp \\ &= \left[\begin{array}{c|c} A_0 & B_0 \\ \hline E_\perp \hat{C}_{\Pi_2} & E' \end{array} \right] \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline -C_0 & E \end{array} \right] + E'_\perp E_\perp \\ &= \left[\begin{array}{cc|c} A_0 & B_0(-C_0) & B_0 E \\ 0 & A_0 - B_0 C_0 & B_0 E \\ \hline E_\perp \hat{C}_{\Pi_2} & E'(-C_0) & E' E + E'_\perp E_\perp \end{array} \right]. \end{aligned}$$

After similarity transformation and applying (2.3):

$$(E' + E_\perp \hat{\Pi}_2) M_0^{-1} E + E'_\perp E_\perp = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline E_\perp \hat{C}_{\Pi_2} - E' C_0 & I \end{array} \right]$$

Therefore,

$$\Psi_{011} = \left[\begin{array}{c} -\hat{\Pi}_1 M_0^{-1} E \\ (E' + E_\perp \hat{\Pi}_2) M_0^{-1} E + E'_\perp E_\perp \end{array} \right] = \left[\begin{array}{c|c} A_0 - B_0 C_0 & B_0 E \\ \hline -\hat{C}_\Pi & 0 \\ E_\perp \hat{C}_{\Pi_2} - E' C_0 & I \end{array} \right]$$

Applying Notation 2.2, the inverse of $E\tilde{P}$ yields

$$(E\tilde{P})^{-1} = \left(\left[\begin{array}{c|c|c} I & \tilde{A} & B \\ \hline & EC & ED \end{array} \right] \right)^{-1} = \left[\begin{array}{cc|cc|c} I & 0 & \tilde{A} & B & 0 \\ 0 & 0 & -EC & -ED & I \\ \hline & & 0 & I & 0 \end{array} \right].$$

Series interconnection is built in a similar way to that of the standard state-space repre-

sentation. Thus, the first row of Ψ_{012} is

$$\begin{aligned}
 (E\tilde{P})^{-1}M_0^{-1}M_1 &= \left[\begin{array}{cc|cc|c} I & 0 & \tilde{A} & B & 0 \\ 0 & 0 & -EC & -ED & I \end{array} \right] \left[\begin{array}{c|c} A_0 - B_0C_0 & -B_1 \\ \hline -C_0 & 0 \end{array} \right] \\
 &= \left[\begin{array}{ccc|ccc} I & 0 & 0 & A_0 - B_0C_0 & 0 & 0 \\ 0 & I & 0 & 0(-C_0) & \tilde{A} & B \\ 0 & 0 & 0 & I(-C_0) & -EC & -ED \end{array} \right] \\
 &\quad \left[\begin{array}{ccc|ccc} \hline & & & 0(-C_0) & 0 & I \\ & & & 0 & 0 & 0 \end{array} \right] \\
 &= \left[\begin{array}{ccc|ccc} I & 0 & 0 & A_0 - B_0C_0 & 0 & 0 \\ 0 & I & 0 & 0 & \tilde{A} & B \\ 0 & 0 & 0 & -C_0 & -EC & -ED \end{array} \right] \\
 &\quad \left[\begin{array}{ccc|ccc} \hline & & & 0 & 0 & I \\ & & & 0 & 0 & 0 \end{array} \right].
 \end{aligned}$$

The second row is

$$\begin{aligned}
 -\tilde{P}(E\tilde{P})^{-1}M_0^{-1}M_1 &= \left[\begin{array}{c|c} \tilde{A} & B \\ \hline -C & -D \end{array} \right] \left[\begin{array}{ccc|ccc} I & 0 & 0 & A_0 - B_0C_0 & 0 & 0 \\ 0 & I & 0 & 0 & \tilde{A} & B \\ 0 & 0 & 0 & -C_0 & -EC & -ED \end{array} \right] \\
 &\quad \left[\begin{array}{ccc|ccc} \hline & & & 0 & 0 & I \\ & & & 0 & 0 & 0 \end{array} \right] \\
 &= \left[\begin{array}{cccc|cccc} I & 0 & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 \\ 0 & 0 & 0 & 0 & -C_0 & -EC & -ED & 0 \\ 0 & 0 & 0 & I & B_0 & B_0 & BI & \tilde{A} \end{array} \right] \\
 &\quad \left[\begin{array}{cccc|cccc} \hline & & & & 0 & 0 & -DI & -C \\ & & & & 0 & 0 & 0 & -D_0 \end{array} \right] \\
 &= \left[\begin{array}{cccc|cccc} I & 0 & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 \\ 0 & 0 & 0 & 0 & -C_0 & -EC & -ED & 0 \\ 0 & 0 & 0 & I & 0 & 0 & B & \tilde{A} \end{array} \right] \\
 &\quad \left[\begin{array}{cccc|cccc} \hline & & & & 0 & 0 & -D & -C \\ & & & & 0 & 0 & 0 & 0 \end{array} \right].
 \end{aligned}$$

Therefore,

$$\Psi_{012} = \begin{bmatrix} (E\tilde{P})^{-1}M_0^{-1}M_1 \\ -\tilde{P}(E\tilde{P})^{-1}M_0^{-1}M_1 \end{bmatrix} =$$

$$\left[\begin{array}{ccc|cccc|c} I & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & -B_1 \\ 0 & I & 0 & 0 & \tilde{A} & B & 0 \\ 0 & 0 & 0 & -C_0 & -EC & -ED & 0 \\ & I & 0 & 0 & 0 & 0 & -B_1 \\ & 0 & I & 0 & 0 & 0 & 0 \\ & 0 & 0 & 0 & -C_0 & -EC & -ED & 0 \\ & 0 & 0 & 0 & 0 & 0 & B & \tilde{A} & 0 \\ \hline & & & 0 & 0 & I & 0 & 0 & 0 \\ & & & & & & 0 & 0 & -D & -C & 0 \end{array} \right]$$

that can be simplified to

$$\Psi_{012} = \left[\begin{array}{cccc|cccc|c} I & 0 & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & 0 & -B_1 \\ 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 & 0 \\ 0 & 0 & 0 & 0 & -C_0 & -EC & -ED & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & B & \tilde{A} & 0 \\ \hline & & & & 0 & 0 & I & 0 & 0 \\ & & & & 0 & 0 & -D & -C & 0 \end{array} \right].$$

For further transformation from descriptor state space to standard state space representation, it is convenient to move the algebraic term to the end (from 3rd to 4th).

$$\Psi_{012} = \left[\begin{array}{cccc|cccc|c} I & 0 & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & 0 & -B_1 \\ 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 \\ 0 & 0 & 0 & 0 & -C_0 & -EC & 0 & -ED & 0 \\ \hline & & & & 0 & 0 & 0 & I & 0 \\ & & & & 0 & 0 & -C & -D & 0 \end{array} \right]$$

After simplification and applying Assumption 2.4 ($ED = 0$)

$$\Psi_{0_{12}} = \left[\begin{array}{ccc|ccc|c} I & 0 & 0 & A_0 - B_0 C_0 & 0 & 0 & -B_1 \\ 0 & I & 0 & 0 & \tilde{A} & B & 0 \\ 0 & 0 & 0 & -C_0 & -EC & 0 & 0 \\ \hline & & & 0 & 0 & I & 0 \\ & & & 0 & -C & -D & 0 \end{array} \right]$$

Based on Assumption 2.6, the relative degree of EP is at most 2, which similarly applies to $E\tilde{P}$. Moreover, the relative degree of M_1 is 2 (for harmonic signals), and the relative degree of M_0 is 0 (see (1.21)). Consequently, $\Psi_{0_{12}}$ is proper, as can be readily seen from the definition of $\Psi_{0_{12}}$ in the transfer function form (2.39). Therefore, $\Psi_{0_{12}}$ may also be converted to standard state-space representation. This problem is well-known, with numerous proven methods ranging from the classical 'shuffle algorithm' [27] and the extraction of algebraic dependencies via singular value decomposition [31] to the extended shuffle algorithm [13] and a recently introduced decomposition method [23]. However, this is outside the scope of this work, and here Ψ_0 will be defined in descriptor state-space representation.

Now append $\Psi_{0_{11}}$ and $\Psi_{0_{12}}$:

$$\left[\begin{array}{cc} \Psi_{0_{11}} & \Psi_{0_{12}} \end{array} \right] = \left[\begin{array}{cccc|cccc|cc} I & 0 & 0 & 0 & A_0 - B_0 C_0 & 0 & 0 & 0 & B_0 E & 0 \\ 0 & I & 0 & 0 & 0 & A_0 - B_0 C_0 & 0 & 0 & 0 & -B_1 \\ 0 & 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -C_0 & -EC & 0 & 0 & 0 \\ \hline & & & & -\hat{C}_\Pi & 0 & 0 & I & 0 & 0 \\ & & & & E_\perp \hat{C}_{\Pi_2} - E' C_0 & 0 & -C & -D & I & 0 \end{array} \right]$$

Finally, gather all the elements of Ψ_0 :

$$\begin{bmatrix} \Psi_{011} & \Psi_{012} \\ \Psi_{021} & \Psi_{022} \end{bmatrix} = \left[\begin{array}{cccccc|cccc} I & 0 & 0 & 0 & 0 & 0 & B_0E & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 & -B_1 \\ 0 & 0 & I & 0 & 0 & \tilde{A} & B & 0 \\ 0 & 0 & 0 & 0 & 0 & -C_0 & -EC & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & A_0 - B_0C_0 & B_0E & B_1 \\ \hline & & & & -\hat{C}_\Pi & 0 & 0 & I \\ & & & & E_\perp \hat{C}_{\Pi_2} - E'C_0 & 0 & -C & -D \\ & & & & 0 & 0 & 0 & -C_0 \end{array} \right] \begin{array}{cc} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}$$

Move the last state to the first:

$$\begin{bmatrix} \Psi_{011} & \Psi_{012} \\ \Psi_{021} & \Psi_{022} \end{bmatrix} = \left[\begin{array}{cccccc|cccc} I & 0 & 0 & 0 & 0 & 0 & B_0E & B_1 \\ 0 & I & 0 & 0 & 0 & 0 & B_0E & 0 \\ 0 & 0 & I & 0 & 0 & A_0 - B_0C_0 & 0 & -B_1 \\ 0 & 0 & 0 & I & 0 & 0 & \tilde{A} & B \\ 0 & 0 & 0 & 0 & 0 & -C_0 & -EC & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline & & & & 0 & -\hat{C}_\Pi & 0 & I \\ & & & & 0 & E_\perp \hat{C}_{\Pi_2} - E'C_0 & 0 & -C \\ & & & & -C_0 & 0 & 0 & 0 \end{array} \right] \begin{array}{cc} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}$$

and simplify it to

$$\Psi_0 = \left[\begin{array}{cccc|cccc} I & 0 & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & 0 & B_0E & 0 \\ 0 & I & 0 & 0 & 0 & A_0 - B_0C_0 & 0 & 0 & 0 & -B_1 \\ 0 & 0 & I & 0 & 0 & 0 & \tilde{A} & B & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -C_0 & -EC & 0 & 0 & 0 \\ \hline & & & & -\hat{C}_\Pi & 0 & 0 & I & 0 & 0 \\ & & & & E_\perp \hat{C}_{\Pi_2} - E'C_0 & 0 & -C & -D & I & 0 \\ & & & & -C_0 & C_0 & 0 & 0 & E & 0 \end{array} \right] \quad (2.43)$$

From (2.43), it is evident that the descriptor state-space representation control comprises the compensator states $2n_m + n$, which determines its order, where n_m represents the

order of the internal model and n denotes the order of the original plant, along with an additional p_e algebraic constraints. This applies to the controller for harmonic signal, while it is less trivial for general repetitive control, and for this moment only a scheme for the SISO system was developed [29].

2.3. IM parameters adaptation for output regulation

From Figure 2.3, it can be seen that

$$\eta = \hat{\Theta}\zeta,$$

where $\zeta = Me$ that can be derived from the third row of (2.37). Moreover, both ζ and e are measurable. Under the following assumption, an adaptation scheme can be designed.

Assumption 2.7.

$$\exists \Theta = \bar{\Theta} \text{ in (2.19) such that } e = \bar{e} \in \mathcal{L}_2.$$

Define the function $z : \mathbb{R}_+ \rightarrow \mathbb{R}^{n_m \times v}$ that satisfies

$$\dot{z}(t) = A_0 z(t) + B_1 \zeta(t), \quad z(0) = 0. \quad (2.44)$$

Define the measurable signal e_1 :

$$e_1 = e + C_0 z \hat{\Theta} - M_1 \hat{\Theta} \zeta,$$

which can be used to apply an adaptation law. For example, the normalised least-squares estimation algorithm [20] is

$$\dot{\hat{\Theta}} = -\frac{\Pi(C_0 z)' e_1}{N} + Q, \quad (2.45)$$

where $Q \geq 0$ (with $Q = 0$ corresponding to the pure least-squares approach), Π is bounded while solving

$$\dot{\Pi} = -\frac{\Pi(C_0 z)'(C_0 z)\Pi}{N} + 2\alpha \frac{\Pi}{N}, \quad \Pi(0) = k_0 I > 0, \quad (2.46)$$

with $\alpha > 0$, $k_0 > 0$, and the scalars

$$N = c + (C_0 z)(C_0 z)' + (C_0 z)\Pi(C_0 z)', \quad c > 0. \quad (2.47)$$

The algorithm's analysis is derived from the choice of the non-negative function

$$V = \tilde{\Theta}' \Pi^{-1} \tilde{\Theta}.$$

This choice implies that $\tilde{\Theta} = \hat{\Theta} - \bar{\Theta}$ is bounded and $\tilde{\Theta} \in \mathcal{L}_2$ (refer to [14]).

3 | Grid-connected inverter

The system being studied is a grid-connected voltage source inverter (VSI) with a loop for controlling the current. The control structure is depicted for a single-phase system (see Figure 3.1); however, the same principles apply to three-phase inverters by utilising identical controllers for each phase in the natural abc frame.

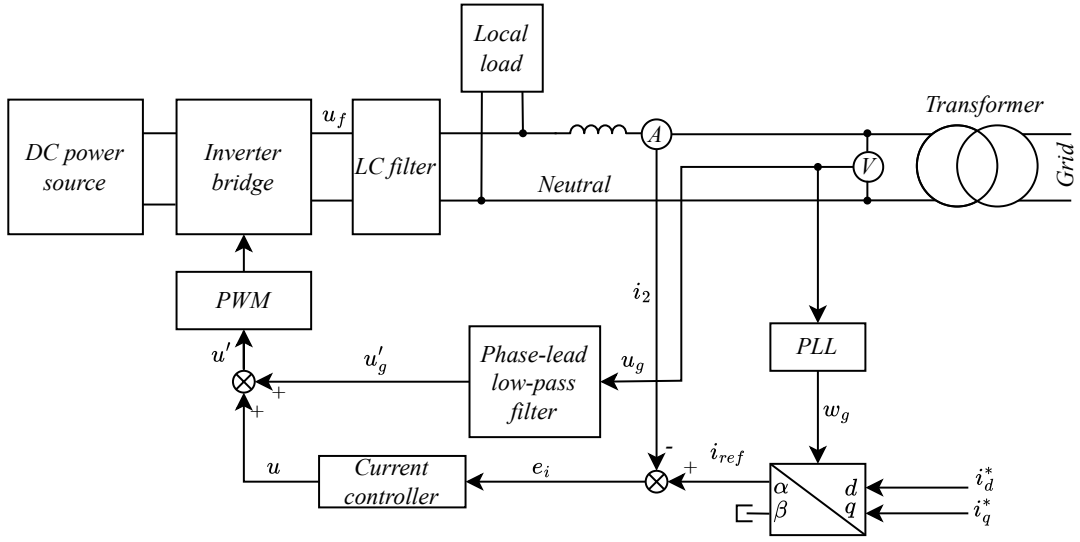


Figure 3.1: Block diagram of a single-phase current-controlled VSI.

The system comprises a DC power source, an inverter bridge utilising pulse width modulation (PWM), an output LC filter, a grid-side L filter, grid synchronisation via a phase-locked loop (PLL), reference generation through Park transformations, and a current controller with feedforward compensation.

A stiff DC power supply powers the inverter, which is assumed to provide a stable DC-link voltage. If this is true, then an external DC-link voltage control loop is unnecessary, and the control design neglects the dynamics on the DC side. This enables the design of the current controller independently of the DC-link voltage regulator.

The inverter bridge is a voltage source inverter comprising semiconductor switching devices such as IGBTs or MOSFETs. By appropriately switching, the bridge converts the DC-link voltage into AC voltage. From a control perspective, the inverter bridge is represented as an ideal voltage source, with its output voltage directly linked to the modulation signal generated by the PWM block.

The pulse-width modulation (PWM) block is configured so that when $|u(t)| < \frac{V_{DC}}{2}$, the average bridge output voltage u_f throughout a switching time is equal to u' . This enables the modelling of the PWM block and the inverter using an average voltage approach [41]. The PWM and inverter model is a straightforward saturated unity gain, whereby the saturation represents the constraint of the available DC-link voltage compared to the neutral line ($\pm \frac{V_{DC}}{2}$).

Due to the operation of an inverter with PWM signals that include harmonics at multiples of the switching frequency, it is essential to connect a low-pass filter to the output of the inverter's power switches to eliminate the harmonics and restore the desired output voltage. This can be performed using traditional passive filters [6]. The grid interface inductor, after the local load, separates the sinusoidal microgrid voltage from possibly distorted grid voltage and facilitates the control of the real and reactive power exchanged between the microgrid and the grid [18]. The LC filter, together with the grid-side inductance, forms an LCL filter between the inverter bridge and the grid. The filter dynamics is very important for keeping the current loop stable, especially at high frequencies where filter resonance is located. The grid is modelled as a perfect voltage source in series with an impedance that represents the grid-side resistance and inductance. Changes in the impedance of the grid affect how the whole system works; that is a significant issue in weak grids [26].

The Phase-locked Loop (PLL) is used to provide the phase information of the grid voltage. This information is needed to ensure that the inverter works with the grid and to create current references that match the grid voltage. The PLL takes the measured grid voltages and gives an estimate of the grid phase. This estimate is considered to be accurate enough for the bandwidth of interest. Concerning single phase PLL, it is important to highlight that the two orthogonal components must be derived from the single phase u to subsequently obtain the DC components V_d and V_q through the application of Park's transformation. The most straightforward method is to employ a transport delay block or a lag compensator to shift the input signal by -90 degrees. This approach is highly sensitive to frequency changes and harmonics, resulting in polluted DC components. In this regard, the second-order generalised integrator (SOGI) should be implemented to provide the PLL with orthogonal components that are filtered and contain only the fundamental

component [7].

In the synchronously rotating reference frame dq [5], the current references are called I_d^* and I_q^* . The d-axis reference determines the active power exchanged with the grid, while the q-axis reference determines the reactive power. The PLL's estimated grid angle is used to change the dq current references into phase current references i_{ref} in the abc frame (or $\alpha\beta$ for single-phase implementations). This ensures that the currents injected are in sync with the grid voltage.

There are many ways to design a current controller [18]. The proportional-integral (PI) controller is widely applied in the industry and is normally designed in the dq frame [36]. A particular advantage of the scheme is the independent control of the real and reactive components of the grid current, which can be directly translated into real and reactive power, making it possible to set the real power and reactive power sent to the grid directly [39]. The PI controller utilised in the dq frame is conventionally regarded as an effective method for regulating sinusoidal currents in balanced three-phase systems. However, it is ineffective at rectifying unbalanced disturbance currents unless the positive and negative sequence currents are addressed independently. Furthermore, the compensating effects of low-order harmonics is insufficient. The PI controller in the synchronously rotating reference frame is more complex than the proportional-resonant (PR) controller in the stationary frame since it necessitates two coordinate transformations and knowledge of the phase information of the grid voltage [18].

The PR controllers can be implemented in stationary and natural reference frames. This allows one to create independent controllers for each phase, improving tracking performance. Moreover, it is possible to add a harmonic compensator in order to improve the power quality [2]. These are the advantages of the PR controller, which make the scheme one of the most popular in grid-connected inverters. The controller has a repetitive or resonant control action that ensures there is no steady-state error when there are periodic references and disturbances, such as the fundamental grid frequency and its harmonics. The proportional part of the controller mostly sets the system's bandwidth, phase margin, and gain margin. The resonant or repetitive part ensures that sinusoidal signals are tracked correctly.

The repeated control method is an alternative approach to provide a clean and balanced current to the grid under grid distortion [17]. The controller can eliminate all periodic disturbances that have the same period as the grid voltage due to its infinite-dimensional internal model. Moreover, it can also be used in PLL to eliminate the unbalanced effect in three-phase inverters [37]. This method is commonly referred to as H^∞ repetitive

control [40], highlighting the primary limitation of the technique: the design of a stabiliser for an infinite-dimensional system is complex. Traditionally, stabilisation is managed by conservative robust control techniques.

The procedure proposed in [8, 9, 20] and studied in Chapter 1 represents an ideal enhancement to traditional PR controller design techniques for grid-connected inverters and constitutes the foundation of this study. Furthermore, this type of system aligns with the assumptions [29], thus enabling the design and discussion of less conservative repetitive control.

A phase-lead low-pass filter $F(s)$ sends the measured grid voltages forward and adds them to the controller output. This makes it possible for quick synchronisation and a smooth grid connection. This filter compensates for the phase lag and gain loss that the PWM, inverter bridge, and output filter cause, especially at the fundamental frequency. So, when the current reference is zero, the inverter output voltage is very close to the grid voltage, which allows the circuit breaker to close without causing transient currents.

When I_d^* and I_q^* are both zero, the inverter works in synchronisation mode and outputs a voltage that is the same as the grid voltage. This is achieved by feed-forwarding the grid voltage and adding it to the output of the current controller via a phase-lead low-pass filter

$$F(s) = \frac{33(0.05s + 1)}{(s + 300)(0.002s + 1)}.$$

It has a gain slightly greater than one and a phase lead at the fundamental frequency. It is implemented to mitigate the phase shift and gain attenuation resulting from computational delay, PWM modulation, the inverter bridge, and the LC filter. It also mitigates the harmonics in the feed-forwarded grid voltages and enhances the dynamics during grid voltage fluctuations [36]. The filter and all system parameters are taken from [18] and are reported here in Table 3.2.

When the inverter produces power, the current controller provides appropriate contributions in addition to the feed-forward grid voltages. Despite the controller being a current controller, the inverter's output remains approximately at the grid voltage level. Therefore, local loads may be connected. When the grid is down, the grid voltage supplied to the filter may be substituted with the rated voltage, and the current references I_d^* and I_q^* can be set to zero. This enables the local loads to function well, although in an open-loop configuration.

3.1. Model of the control plant

Figure 3.2 illustrates the system's diagram. The system comprises the inverter bridge, an inverter-side LC filter (L_1 and C), and a grid-interface inductor L_2 . The model incorporates the inverter-side and grid-side inductances, including their series parasitic resistances to account for winding resistance while neglecting core losses; i.e., no parallel loss resistance is included. R_d denotes the equivalent series resistance of the filter capacitor. The resistances facilitate the dampening of the LC resonance and may thus be intentionally increased, although this can lead to significant power losses [34]. When an inverter is connected to a weak grid, grid inductance and resistance can be included (L_g and R_g).

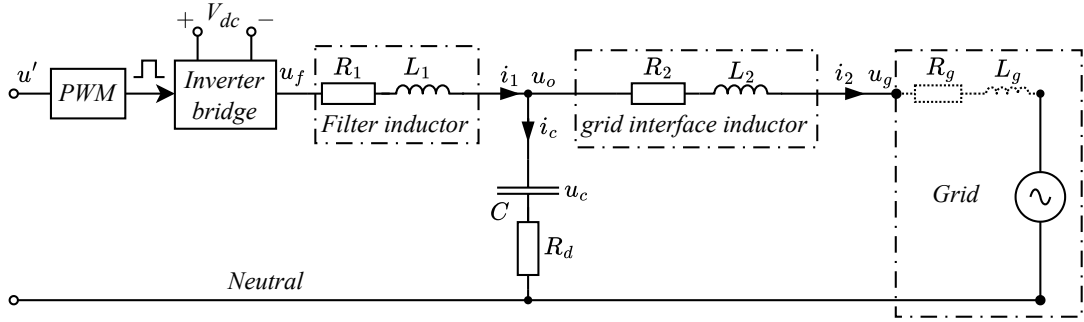


Figure 3.2: Representation of the plant.

The control input is the inverter voltage $u' = u + u'_g$, where the feed-forwarded grid voltage u'_g serves as the reference output voltage, which is present on both sides of the LCL filter. It does not influence the controller design and can be disregarded during the design process. The sole factor to be considered during the design phase is the output u of the repetitive current controller. The system has the following impedances:

$$Z_f = L_1 s + R_1 \quad (3.1)$$

$$Z_g = (L_2 + L_g)s + R_2 + R_g \quad (3.2)$$

$$Z_0 = \frac{1}{C_s} + R_d \quad (3.3)$$

From the scheme, the impedance Z_f is in series with a parallel connection of Z_g and Z_0 ; thus, the total impedance seen by u_f is

$$Z_{tot} = Z_f + \frac{Z_0 Z_g}{Z_0 + Z_g} = \frac{Z_1 Z_0 + Z_1 Z_g + Z_g Z_0}{Z_0 + Z_g}.$$

By Ohm's law,

$$i_1 = \frac{u_f}{Z_{tot}}.$$

Substituting the expression of Z_{tot} yields

$$G_{u_f-i_1}(s) = \frac{i_1(s)}{u_f(s)} = \frac{Z_g + Z_0}{Z_i Z_g + Z_i Z_0 + Z_g Z_0}. \quad (3.4)$$

Currents in the parallel connection are divided according to impedances:

$$i_2 = i_1 \frac{Z_0}{Z_g + Z_0}$$

Thus,

$$G_{i_1-i_2}(s) = \frac{i_2(s)}{i_1(s)} = \frac{Z_0}{Z_g + Z_0}. \quad (3.5)$$

Combine (3.5) with (3.4)

$$G_{u_f-i_2}(s) = \frac{i_2(s)}{i_1(s)} \frac{i_1(s)}{u_f(s)} = \frac{Z_0}{Z_g + Z_0} \frac{Z_g + Z_0}{Z_i Z_g + Z_i Z_0 + Z_g Z_0}.$$

Cancel $Z_g + Z_0$:

$$G_{u_f-i_2}(s) = \frac{Z_0}{Z_i Z_g + Z_i Z_0 + Z_g Z_0} \quad (3.6)$$

Define $L = L_1 + L_2 + L_g$, $R_{2g} = R_2 + R_g$, and $L_{2g} = L_2 + L_g$. Substituting equations (3.1)–(3.3) in (3.4)–(3.6), one gets

$$G_{u_f-i_1}(s) = \frac{CL_{2g}s^2 + C(R_d + R_{2g})s + 1}{\left(\begin{array}{l} CL_1L_{2g}s^3 + C(L_1R_{2g} + R_1L_{2g} + R_dL)s^2 \\ + (L + C[R_1(R_d + R_{2g}) + R_dR_{2g}])s + R_1 + R_{2g} \end{array} \right)}, \quad (3.7)$$

$$G_{i_1-i_2}(s) = \frac{CR_d s + 1}{C(L_2 + L_g)s^2 + C(R_2 + R_d + R_g)s + 1}, \quad (3.8)$$

$$G_{u_f-i_2}(s) = \frac{CR_d s + 1}{\left(\begin{array}{l} CL_1L_{2g}s^3 + C(L_1R_{2g} + R_1L_{2g} + R_dL)s^2 \\ + (L + C[R_1(R_d + R_{2g}) + R_dR_{2g}])s + R_1 + R_{2g} \end{array} \right)}. \quad (3.9)$$

The transfer function that will mostly be used is (3.9).

Now, derive the state space representation of the plant. Differential equations for electrical elements [11] are represented in Table 3.1.

Physical Element	Governing Equation
Electrical inductance	$u = L \frac{di}{dt}$
Electrical capacitance	$i = C \frac{du}{dt}$
Electrical resistance	$i = \frac{1}{R} u$

Table 3.1: Differential Equations for Electrical Elements

The state vector is chosen as

$$x = \begin{bmatrix} i_1 & i_2 & u_c \end{bmatrix}^T,$$

where i_1 is the inverter-side inductor current, i_2 is the grid-side inductor current (the current injected into the grid), and u_c is the filter capacitor voltage. The disturbance input is the grid voltage $w = u_g$, and the control input is u . The regulated output is i_2 ; depending on the control design, i_1 may also be measured and used for stabilisation/performance shaping (e.g., to improve damping of the LCL resonance [35]). Using the expressions in Table 3.1, Kirchhoff's voltage law for the inverter-side and grid-side inductor loops yields:

$$\begin{aligned} L_1 \dot{i}_1 &= u_f - R_1 i_1 - u_0 = u + u_g - R_1 i_1 - u_c - R_d(i_1 - i_2) \\ L_2 \dot{i}_2 &= u_o - R_2 i_2 - L_g \dot{i}_2 - R_g i_2 = u_c + R_d(i_1 - i_2) - R_2 i_2 - L_g \dot{i}_2 - R_g i_2 - u_g \end{aligned}$$

Kirchhoff's Current Law at the capacitor node

$$C \dot{u}_c = i_1 - i_2.$$

Differential equations for the inverter states follow from the written equations through straightforward algebra:

$$\dot{i}_1 = -\frac{R_1 + R_d}{L_1} i_1 + \frac{R_d}{L_1} i_2 - \frac{1}{L_1} u_c + \frac{1}{L_1} u_g + \frac{1}{L_1} u \quad (3.10)$$

$$\dot{i}_2 = \frac{R_d}{L_{2g}} i_1 - \frac{R_d + R_{2g}}{L_{2g}} i_2 + \frac{1}{L_{2g}} u_c - \frac{1}{L_{2g}} u_g \quad (3.11)$$

$$\dot{u}_c = \frac{1}{C} i_1 - \frac{1}{C} i_2 \quad (3.12)$$

The state equation is

$$\dot{x} = Ax + B_1 w + B_2 u, \quad (3.13)$$

and the output equation is

$$y = Cx + D_1w + D_2u \quad (3.14)$$

and, using Notation 1.1, the plant can be written as

$$P = \left[\begin{array}{c|c|c} A & B_1 & B_2 \\ \hline C & D_1 & D_2 \end{array} \right] = \left[\begin{array}{ccc|ccc} -\frac{R_1+R_d}{L_1} & \frac{R_d}{L_1} & -\frac{1}{L_1} & \frac{1}{L_1} & \frac{1}{L_1} & \\ \frac{R_d}{L_{2g}} & -\frac{R_d+R_{2g}}{L_{2g}} & \frac{1}{L_{2g}} & -\frac{1}{L_{2g}} & 0 & \\ \frac{1}{C} & -\frac{1}{C} & 0 & 0 & 0 & \\ \hline 0 & 1 & 0 & 0 & 0 & \end{array} \right]. \quad (3.15)$$

When the inverter-side current i_1 is also measured (used for stabilisation/performance shaping), the measured output is $y = \begin{bmatrix} i_2 & i_1 \end{bmatrix}^T$ with

$$C = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

3.2. System analysis

The inverter parameters are given in Table 3.2. For further comparison of the proposed control scheme with a conventional design, the parameters are taken from [18].

Parameter	Value	Parameter	Value
L_1	$150\mu H$	L_2	$450\mu H$
R_1	0.045Ω	R_2	0.135Ω
C	$22\mu F$	R_g	0Ω
R_d	1Ω	L_g	0Ω
V_{DC}	$40V$	u_g	$15V, 50Hz$
f_{sw}	$12kHz$	f_s	$5kHz$

Table 3.2: Parameters of the inverter for simulation

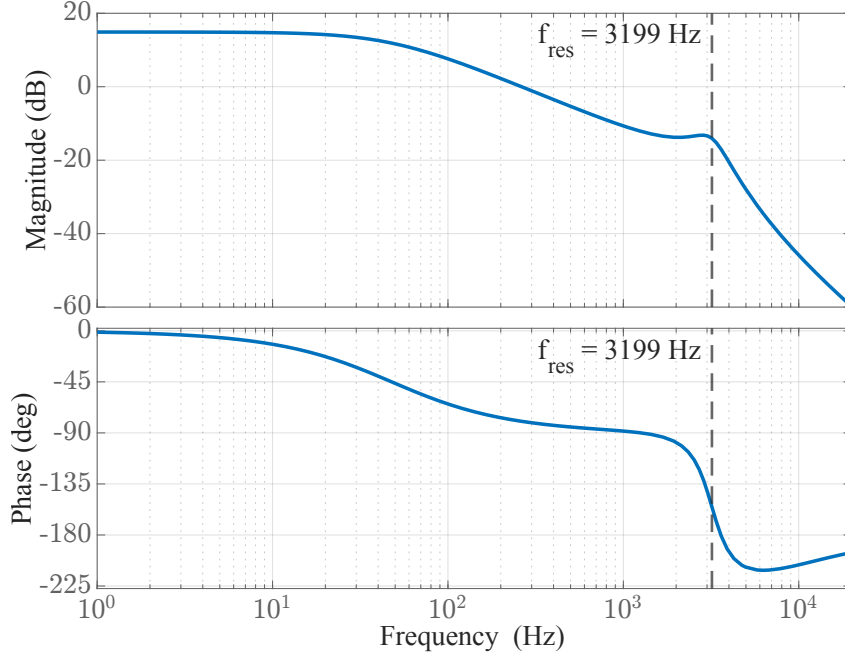


Figure 3.3: Bode plot of the LCL filter transfer function from u_f to i_2

Figure 3.3 illustrates the Bode diagram of the LCL filter's transfer function. At low frequencies ($\leq 50\text{Hz}$), the magnitude response is nearly flat, and the phase is small, meaning that the inverter output voltage is efficiently delivered to the grid current with minimal attenuation or phase shift. This frequency range pertains to the fundamental grid frequency, which is the primary control target of the current controller.

As frequency rises, magnitude progressively decreases and phase transitions towards -90° , indicating the inductive characteristics of the filter. At the LCL resonance frequency f_{res} , a significant resonance peak emerges in the magnitude response, accompanied by a rapid phase decline beyond -180° . This resonance results from the energy transfer between the filter inductors and the filter capacitor.

Beyond the resonance frequency, the magnitude response displays significant attenuation, illustrating the effectiveness of the LCL filter in reducing high-frequency components, including switching harmonics produced by the PWM inverter. The phase progressively diminishes and nears -180° at high frequencies, which significantly affects closed-loop stability.

The existence of the resonance peak and the corresponding phase lag imposes strict constraints on the attainable control bandwidth and requires the implementation of suitable damping methods to guarantee the stable functioning of the grid-connected inverter. Passive damping is commonly used for preserving system stability; however, it is constrained

by losses, expenses, and decay of filter efficiency [4]. Active damping may provide a solution [26], but it requires a sophisticated tuning procedure for the controllers [3, 25, 28]. The specific feedback method can be applied with a weighted average of inverter-side and grid currents, effectively eliminating LCL resonance in the control loop [32]. This enables increased proportional gain and broader bandwidth for PR harmonic control; however, it necessitates an additional current sensor and precise weighting depending on grid inductance.

Figure 3.4 illustrates the influence of passive damping on the characteristics of the filter at the resonance frequency and in the high-frequency region. In the absence of damping, the response declines at -60 dB per decade; yet, the resonance peak induces instability in the system. Increasing damping resistance diminishes the resonance spike; nevertheless, the attenuation capability is also reduced to -40 dB/decade with $R_d = 1\Omega$, leading to an increase in filter size. Nonetheless, even with 1Ω damping, the LCL filter demonstrates superior efficiency compared to the LC filter (see Figure 3.5), necessitating a significant increase in the inductance and capacitance of the latter to achieve equivalent attenuation capabilities.

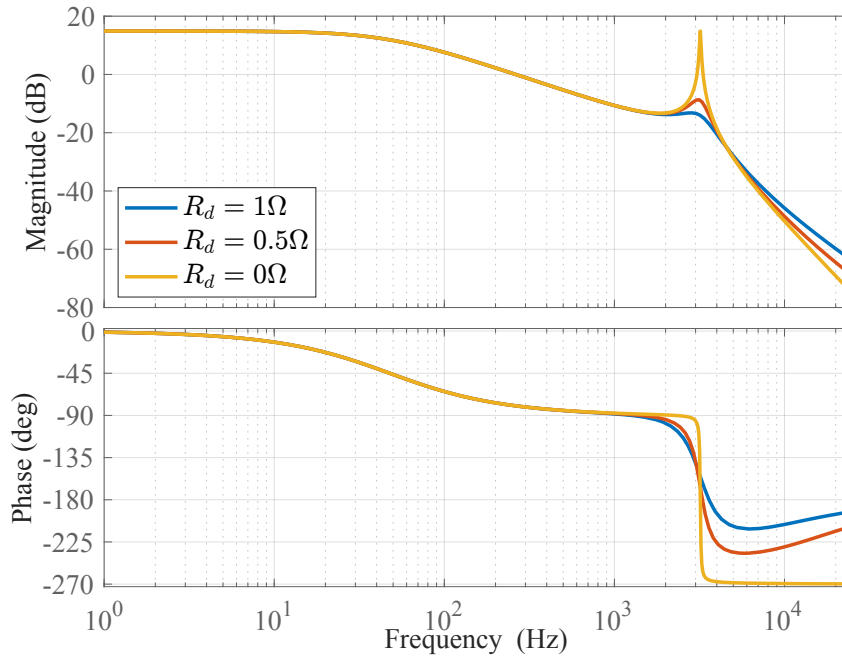


Figure 3.4: Bode plots of the current transfer function of the damped LCL filters.

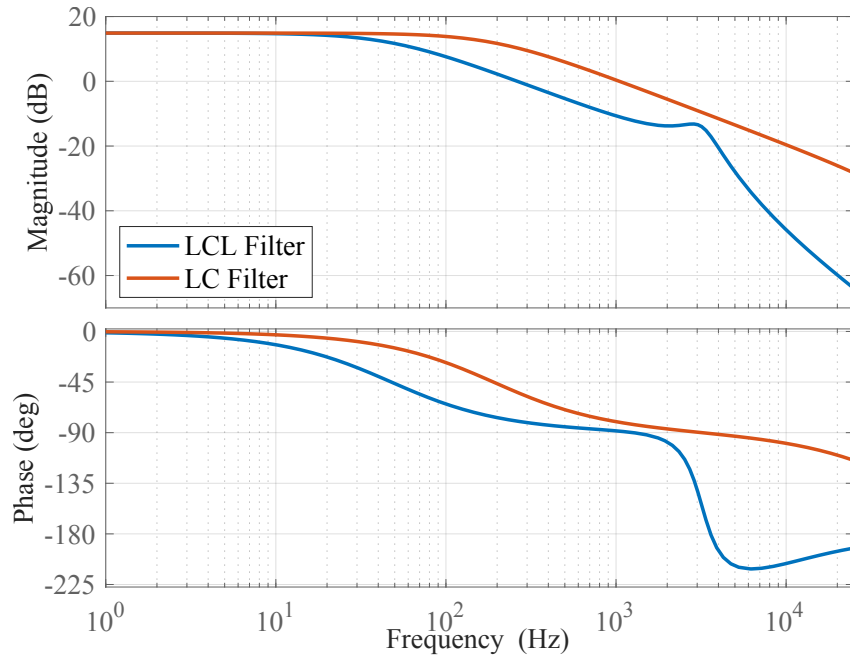


Figure 3.5: Frequency characteristics of the LCL- and LC-filters. All the parameters are the same, except for $L_2 = 0\mu H$ and $R_d = 0\Omega$ for latter.

4 | Proposed control scheme for grid-connected inverter

4.1. Current PR-control with CIM

Consider plant (3.15) with parameters from Table 3.2:

$$P = \left[\begin{array}{c|c|c} A & B_1 & B_2 \\ \hline C & D_1 & D_2 \end{array} \right] = \left[\begin{array}{ccc|cc} -\frac{R_1+R_d}{L_1} & \frac{R_d}{L_1} & -\frac{1}{L_1} & \frac{1}{L_1} & \frac{1}{L_1} \\ \frac{R_d}{L_{2g}} & -\frac{R_d+R_{2g}}{L_{2g}} & \frac{1}{L_{2g}} & -\frac{1}{L_{2g}} & 0 \\ \frac{1}{C} & -\frac{1}{C} & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 & 0 \end{array} \right] \quad (4.1)$$

$$= \left[\begin{array}{ccc|cc} -6967 & 6667 & -6667 & 6667 & 6667 \\ 2222 & -2522 & 2222 & -2222 & 0 \\ 45455 & -45455 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 & 0 \end{array} \right]$$

For controller design, the exogenous signal can be omitted. Transfer function representation is

$$P = \frac{1.4815 \times 10^7 (s + 4.545 \times 10^4)}{(s + 300)(s^2 + 9189s + 4.04 \times 10^8)}. \quad (4.2)$$

Consider that the grid is distorted by the 3rd and 5th harmonics. Then there are three frequencies of exogenous signals, including the fundamental:

$$\omega_1 = 2\pi \cdot 50, \quad \omega_2 = 2\pi \cdot 150, \quad \omega_3 = 2\pi \cdot 250$$

In a typical grid, the peaks of its voltage are flattened [18]. As an example (see Figure 4.1), the grid voltage is modelled as

$$u_g = 15 \sin \omega t - 1.5 \sin \omega_2 t - 1.5 \sin \omega_3 t. \quad (4.3)$$

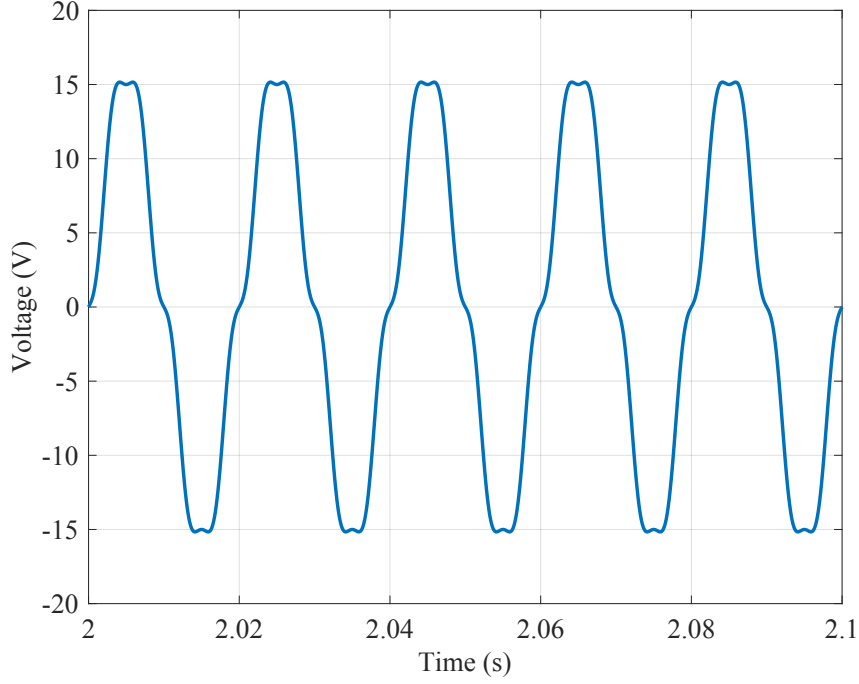


Figure 4.1: Steady-state grid voltage.

Thus, the internal model can be chosen as

$$M(s) = \frac{(s + a_m)^6}{(s^2 + \omega_1^2)(s^2 + \omega_2^2)(s^2 + \omega_3^2)}, \quad (4.4)$$

with $a_m > 0$, satisfying Assumptions 2.1 and 2.3. The system has one output; therefore, matrix $E = 1$. Moreover, $E_\perp = 0$ from (2.3), and only one compensator is used. Thus, R_s from (2.4) is

$$R_s = \bar{R} - \Pi_1 E, \quad (4.5)$$

with Π_1 from (2.11):

$$\Pi_1(s) = \left[\begin{array}{c|c} \frac{A_m - B_m C_m}{C_\Pi} & \frac{B_m}{0} \end{array} \right] \quad (4.6)$$

Given (4.2), $a_m = 80$, and making reference to Proposition 2.1, one has

$$\bar{P} = \frac{1.4815 \times 10^7 (s + 4.545 \times 10^4)}{(s - 2697)(s^2 + 9170s + 4.117 \times 10^8)}, \quad (4.7)$$

and an internal model compensator is

$$\Pi_1 = \frac{-4742.6(s + 300.4)(s^2 + 164.7s + 7.038 \times 10^4)(s^2 + 657.6s + 6.535 \times 10^5)}{(s + 502.7)^6}. \quad (4.8)$$

From the root locus for $\bar{P}$, the stabiliser is chosen as $\bar{R} = -5$ from the range of possible values from -1.65 to -6.2 , while for the original plant, gains in the range from 0 to -7 are stabilising. This shows that any value that stabilises the equivalent plant can stabilise the original plant and can be safely applied. The control scheme is shown in Figure 4.2.

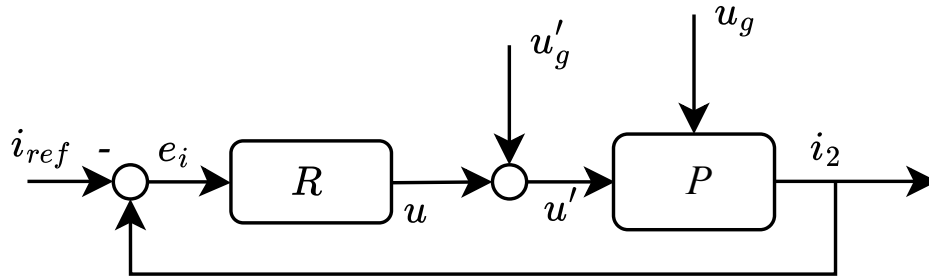


Figure 4.2: Block diagram of the PR-control with CIM.

Simulation results shown in Figure 4.3 demonstrate a strong rejection of the harmonics introduced by flattened voltage in the output current, making the switching frequency the main factor in THD, whose attenuation is not related to the controller but to the LCL-filter design.

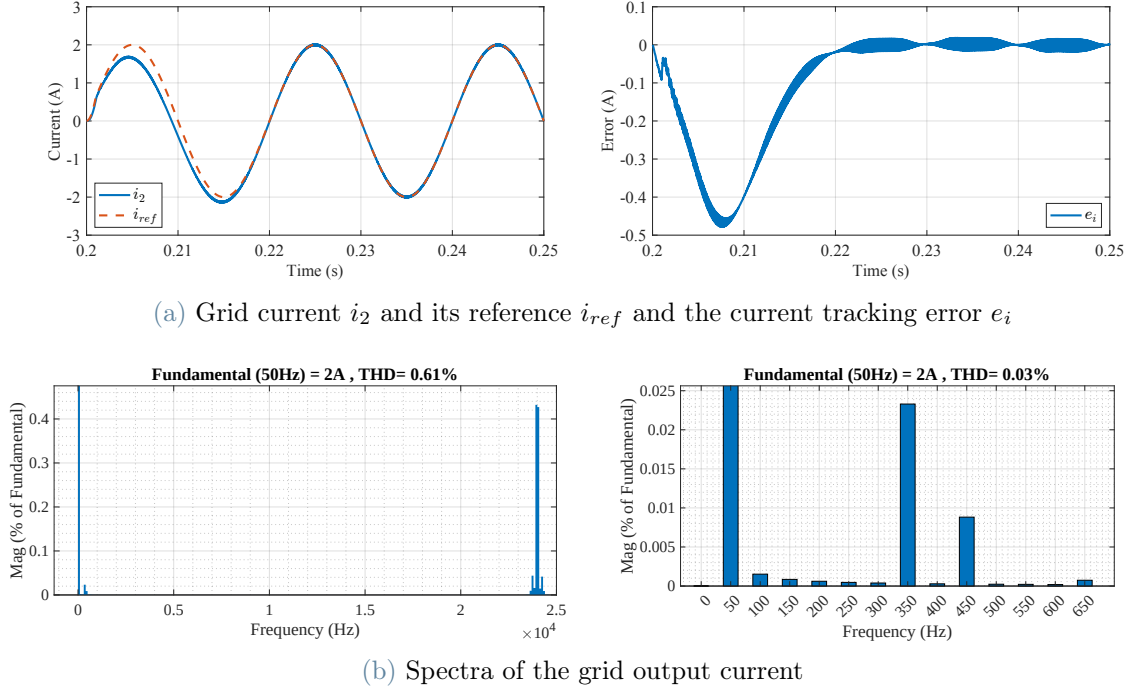


Figure 4.3: Simulation results for current PR-control with CIM.

4.2. Adaptive current PR-control with CIM

Now consider a grid without harmonic distortion; however, with a grid frequency 49.5Hz, while the frequency in the controller is initially set to 50Hz. The internal model is then

$$M(s) = \frac{(s + a)^2}{s^2 + w^2}.$$

Applying factorisation as described in Section 2.2

$$M^{-1}(s) = \frac{s^2}{(s + a)^2} + \frac{w^2}{(s + a)^2} = M_0(s) + M_1(s)\Theta.$$

Equivalent plant and compensator are derived according to Proposition 2.2

$$\begin{aligned} \tilde{P} &= \frac{1.4815 \times 10^7 (s + 4.545 \times 10^4)}{(s - 704.9)(s^2 + 9188s + 4.045 \times 10^8)} \\ \hat{\Pi}_1 &= \frac{-273.96(s + 390.5)}{(s + 502.7)^2} \end{aligned}$$

Apply the pure least squares adaptation law described in Section 2.3 with adaptation parameters $Q = 0$, $\alpha = 100$, $k_0 = 1000$, $c = 0.1$, and the refactored model as described in 2.2. Firstly, consider an ideal case when the inverter switching frequency in the model is

set very high (for example, $f_{sw} = 120000\text{Hz}$), so that its distortion is negligible. Figure 4.4 shows how the estimated frequency perfectly adapts to the real one.

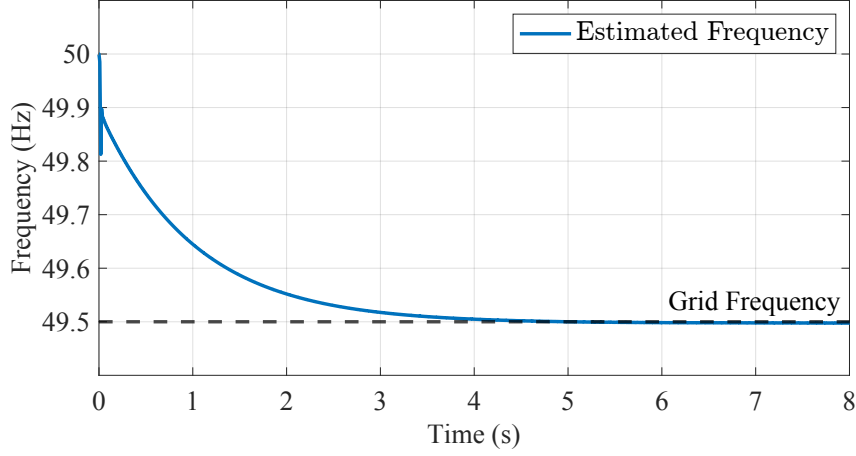
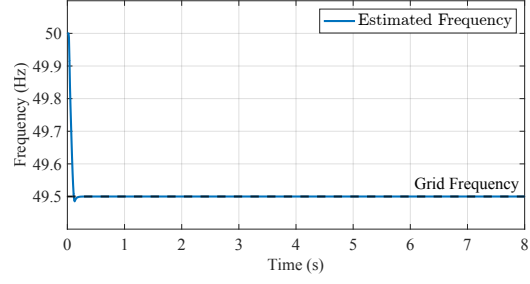
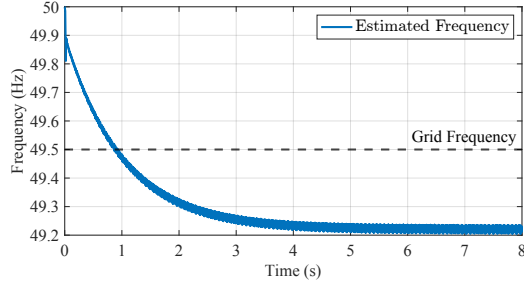
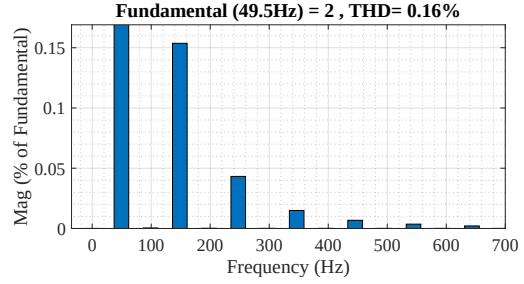
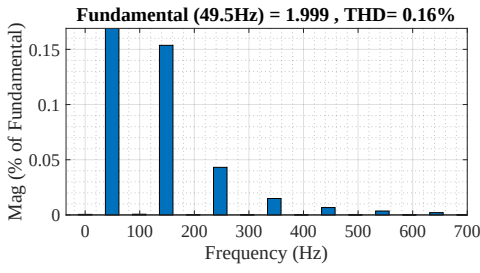


Figure 4.4: Frequency adaptation without inverter bridge distortion

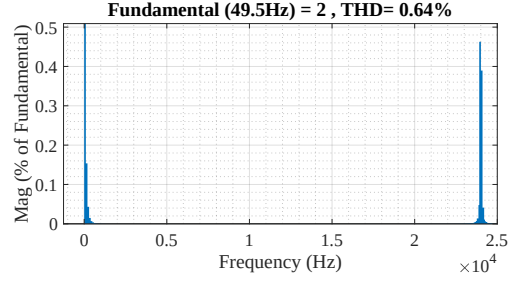
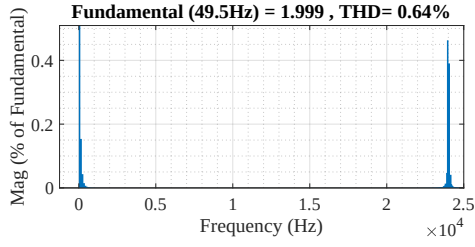
However, when the switching frequency is set to a realistic value of $f_{sw} = 12000\text{Hz}$, as shown in Table 3.2, the estimated frequency is noisy and reaches the value of $49.2...49.25\text{Hz}$. This behaviour is an effect of the control objective, which is minimising current error and not the frequency estimation error. The described adaptation algorithm is compared with a scheme in which the frequency estimated in the phase-locked loop is fed to the controller. It is seen that the error and the total harmonic distortion are almost the same for the two schemes. However, the use of the PLL output frequency is considered a more practical approach for inverter control applications since this element is needed for Park transformation for inverter control, as described in the previous chapter, and the main distortions in the grid are harmonics of the fundamental frequencies.



(a) Frequency adaptation



(b) Spectra of the grid output current in the steady state (low frequencies)



(c) Spectra of the grid output current in the steady state (including switching frequency)

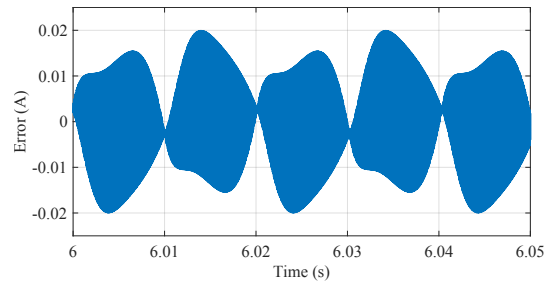
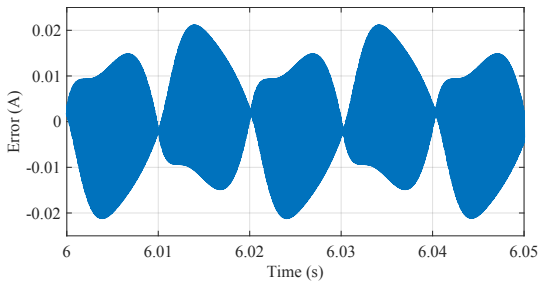
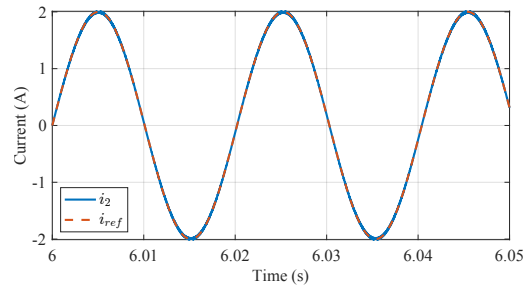
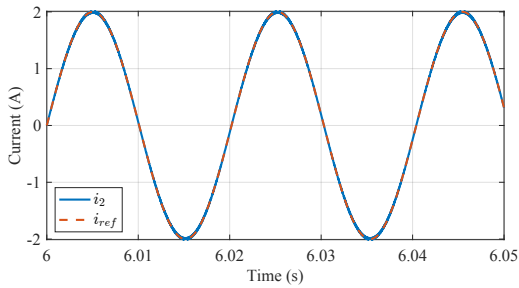
(d) Current tracking error e_i in the steady-state(e) Grid current i_2 and its reference i_{ref} in the steady-state

Figure 4.5: Simulation results for adaptive current PR-control with CIM: least square estimation (left column) and PLL estimation (right column).

4.3. Current PR Control with IM in subset and CIM

Consider the same setup and disturbance model as in Section 4.1. To improve the damping of the LCL filter resonance [35] and keep the stabiliser operational with the same sensing set, whether the grid is connected or not, the inverter-side current i_1 can be measured along with the grid-side current i_2 . The measured output is defined as

$$y = \begin{bmatrix} i_2 \\ i_1 \end{bmatrix} = Cx, \quad C = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

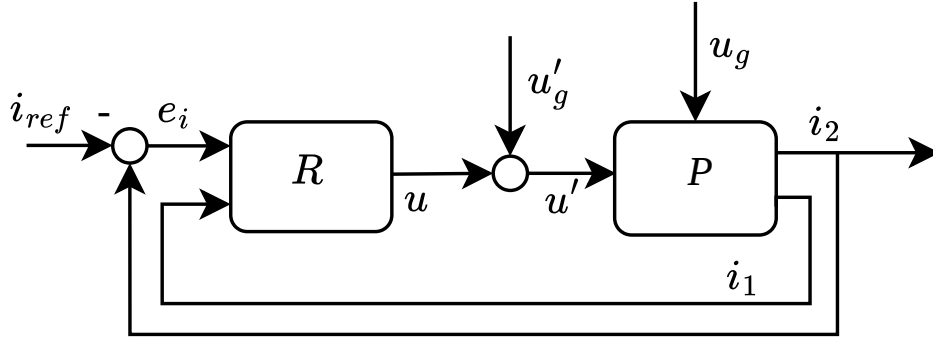


Figure 4.6: Block diagram of the PR-control with IM in subset of channels and CIM.

Regulated output applies solely to the grid current i_2 ($p_e = 1$), while i_1 serves only to enhance performance (see Figure 4.6). Consequently, the internal model is only implemented within the i_2 channel. The selection matrix changes to $E = \begin{bmatrix} 1 & 0 \end{bmatrix}$. Now, in the stabiliser of the form (2.4), two compensators should be used in the controller:

$$R_s = \bar{R}(I + E'_\perp \Pi_2 E) - \Pi_1 E,$$

where $\bar{R}$ is designed for an equivalent plant, which is defined as (Proposition 2.1)

$$\bar{P} = \begin{bmatrix} 1.4815 \times 10^7(s + 4.545 \times 10^4) \\ 6666.7(s^2 - 493.7s + 1.092 \times 10^8) \end{bmatrix} \frac{1}{(s - 2697)(s^2 + 9170s + 4.117 \times 10^8)}.$$

An observer-based optimal stabiliser, such as a Linear Quadratic Gaussian (LQG) or a stabiliser with efficient virtual impedance-based active damping for LCL filter reso-

nance [35], could indeed be formulated; however, this is not the primary focus of the current discussion. The stabiliser is selected as a static output-feedback gain, expressed as $u = Ky$, where $K = -\begin{bmatrix} 5 & 0.1 \end{bmatrix}$. This selection is achieved through an iterative trial-and-error process until a stable closed loop is attained. The closed-loop matrix $\tilde{A} + BKC$ has eigenvalues

$$\lambda(\tilde{A} + BKC) = \begin{bmatrix} -984 + 21202i \\ -984 - 21202i \\ -5171 \end{bmatrix},$$

indicating that the stabiliser utilised for the subsequent CIM-based regulation design demonstrates internal stability for the equivalent plant. Again, by Proposition 2.1, the compensators are

$$\begin{aligned} \Pi_1 &= \frac{-4742.6(s + 300.4)(s^2 + 164.7s + 7.038 \times 10^4)(s^2 + 657.6s + 6.535 \times 10^5)}{(s + 502.7)^6} \\ \Pi_2 &= \frac{3168.8(s - 498.1)(s^2 + 43.84s + 1.208 \times 10^5)(s^2 + 629.6s + 1.043 \times 10^6)}{(s + 502.7)^6} \end{aligned}$$

The simulation results are shown in Figure 4.7.

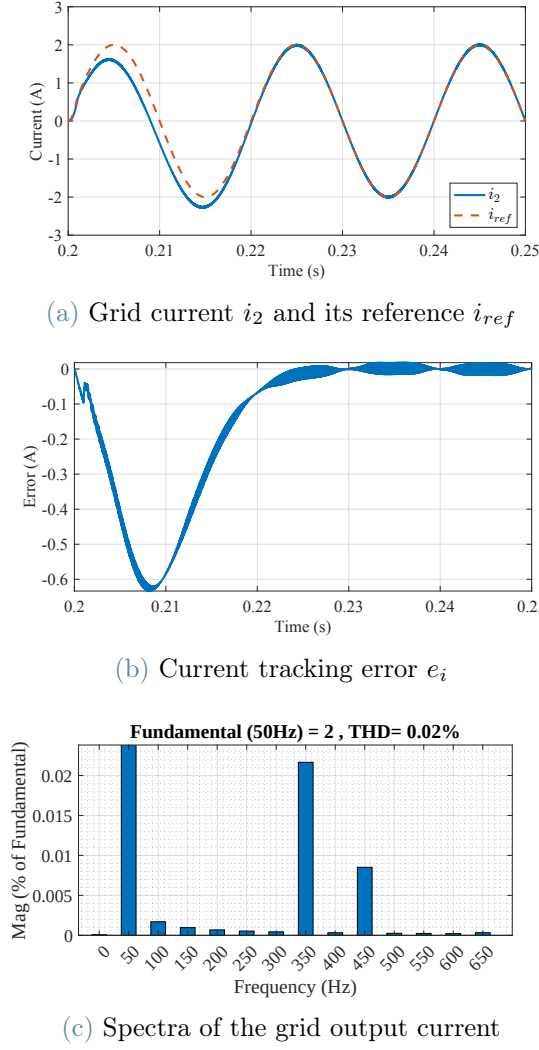


Figure 4.7: Simulation results for current PR-control with IM in a subset of output channels and CIM.

4.4. Design rationale

Consider the approach to designing a stabilising controller through the placement of poles, or more generally, by employing any state-feedback synthesis method, provided that the states of the plant can be measured. In the conventional internal model regulator approach, the plant is augmented with the internal model. Consequently, the stabilisation problem is framed within an enlarged state vector of dimension $n + n_m$, requiring the designer to assign (or optimise) the closed-loop poles of the augmented system. In this manner, the synthesis phase includes both the states of the plant and the states of the internal model.

In the CIM framework, the internal model is presented through a compensation structure,

whereas the stabiliser is developed for an equivalent plant. The stabilisation step operates on a system equivalent plant of order n and, according to the assumptions outlined in Chapter 2, is independent of the parameters of the internal model.

With appropriate tuning, both methods can achieve comparable closed-loop performance on the chosen benchmark. Consequently, this thesis does not aim to demonstrate that one method outperforms the other regarding performance or robustness. The focus is on the design methodology: CIM provides a modular workflow that reduces the complexity and dependence on the stabilisation step when introducing or modifying internal models. This is particularly beneficial when internal-model parameters are tuned or adapted.

From a practical standpoint, a heuristic design approach is widely applied in industrial settings for grid-connected converters [2]. Typically, the stabiliser is first designed for the nominal plant, after which resonant components are incorporated and tuned while monitoring stability margins. In this case study, this pragmatic approach can yield performance that is competitive with, and in some cases better than, the assessed CIM implementations, particularly because the chosen plant is stable. However, the heuristic procedure is not a systematic synthesis method and does not directly provide a structural route to parameter adaptation, whereas the CIM framework is systematic and naturally accommodates adaptation mechanisms through its separation between stabilisation and internal-model embedding.

For repetitive control, a commonly employed systematic synthesis baseline is grounded in H^∞ design for repetitive structures. This approach facilitates straightforward comparisons of results using the same plant model and identical internal model components, specifically the period τ and the low-pass filter. Repetitive control also establishes an internal model characterised by delay-like (infinite-dimensional) properties, thereby highlighting the benefits of compensation and controller structuring more clearly than in resonant controller designs. For these reasons, the following section provides a comparison between CIM-based repetitive control and the standard H^∞ repetitive controller.

4.5. Current repetitive control with CIM

One can notice that P in (4.2) is stably invertible. In this case, repetitive control with CIM may be easily applied. Firstly, define the internal model for the repetitive element as [10]

$$M(s) = \frac{1}{1 - F(s)e^{-\tau s}} I_{pe},$$

where $F(s)$ is a stable finite-dimensional low-pass filter, $\tau > 0$, and I_{p_e} is a $p_e \times p_e$ identity matrix. Again, for current control $p_e = 1$.

Since $M_0 = 1$ the Sylvester equation (2.28)

$$\begin{bmatrix} B^\perp \\ 0 \end{bmatrix} X A_0 - \begin{bmatrix} B^\perp A \\ -EC \end{bmatrix} X = \begin{bmatrix} 0 \\ -C_0 \end{bmatrix}$$

results in

$$\begin{bmatrix} B^\perp A \\ -EC \end{bmatrix} X = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and has a unique solution $X = 0$ if $\begin{bmatrix} B^\perp A \\ -EC \end{bmatrix}$ has full column rank, which is, in fact, the case.

Thus $\tilde{P}$ based on the proposition (2.2) yields

$$\tilde{P}(s) = \left[\frac{A + XB_0EC}{C} \middle| \frac{B}{D} \right] = \left[\frac{A}{C} \middle| \frac{B}{D} \right] = P.$$

CIM elements (2.27) such that no dependence on $e^{-\tau s}$ occurs are

$$\begin{bmatrix} \hat{\Pi}_1 \\ \hat{\Pi}_2 \end{bmatrix} := \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} - \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} M_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Full compensators are

$$\begin{bmatrix} \tilde{\Pi}_1 \\ \tilde{\Pi}_2 \end{bmatrix} := - \begin{bmatrix} I \\ E_\perp P \end{bmatrix} (EP)^{-1} M_1 e^{-\tau s}.$$

For compensators to be proper, the $M_1 = -F(s)$ should have a relative degree of 2, since the relative degree of the P is 2 (see (4.2)). Thus, the second-order Butterworth filter is chosen

$$F(s) = \frac{w_\phi^2}{s^2 + \sqrt{2}w_\phi + w_\phi^2} = \left[\frac{A_F}{C_F} \middle| \frac{B_F}{0} \right], \quad (4.9)$$

where w_ϕ is a tuning parameter on which tracking and disturbance rejection capabilities depend; i.e., the larger w_ϕ , the higher the frequencies that can be effectively tracked or rejected. For the filter above $|1 - F(jw)| < 0.01$ whenever $w < 0.07w_\phi$. For the chosen

$w_\phi = 2\pi \cdot 2550 \text{ rad/s}$, then $w = 0.07 \cdot 2\pi \cdot 2550 = 2\pi \cdot 178,5 \text{ rad/s}$, and good attenuation or reference tracking is expected for frequencies less than 178.5 Hz .

Since $E = 1$, the only compensator Π_1 is used

$$\Pi_1 = -P^{-1}M_1e^{-\tau s}.$$

Regulator (2.21) for the augmented plant (2.2) is

$$\tilde{R}_s = \tilde{R} - \tilde{\Pi}_1 = \tilde{R} + P^{-1}M_1e^{-\tau s}. \quad (4.10)$$

Therefore, applying the equations above and the M^{-1} definition, the controller (2.25) is

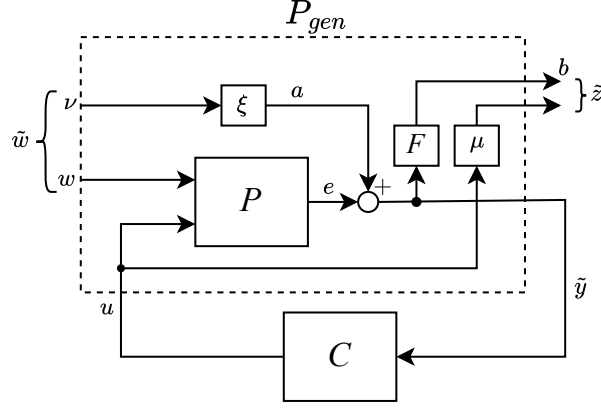
$$R = R_s M = (\tilde{R} + P^{-1}M_1e^{-\tau s})(1 + M_1e^{-\tau s})^{-1}.$$

By Lemma 2.1, one can easily obtain the controller in the form (2.33)

$$R = \mathcal{LFT}_l(\Psi, e^{-\tau s}) = \mathcal{LFT}_l\left(\begin{bmatrix} \tilde{R} & (\tilde{R} - P^{-1})F \\ 1 & F \end{bmatrix}, e^{-\tau s}\right). \quad (4.11)$$

This form allows not only the implementation of the same stabiliser for the system as for the original plant (for example, simple gain) instead of the H^∞ technique, but also allows for slow adaptation. The delay element is stable if $|\dot{\tau}(t)| < 1$ [29]. Since, in terms of frequency delay, $\tau = \frac{1}{f}$ and $\dot{\tau}(t) = -\frac{\dot{f}(t)}{f^2}$ are applicable for the power grid application where $f \approx 50 \text{ Hz}$, the maximum rate of frequency change for stability guarantee is $\dot{f}(t) \lesssim 2500 \text{ Hz/s}$, which is enough for this application since the rate of change of frequency in the grids is much lower [1]. Moreover, the rate of change can be set in the PLL to ensure the stability of the delay element.

Further, the controller (4.11) with regulator $\tilde{R} = -1.1$ (provides a good compromise between harmonic rejection and robustness) is compared to the current H^∞ repetitive controller designed based on the approach in [18] with filter (4.9) in the repetitive element. The formulation of the H^∞ control problem is shown in Figure 4.8.

Figure 4.8: Formulation of the H^∞ control problem

The controller synthesis is framed as a conventional H^∞ optimisation problem to guarantee closed-loop stability and achieve the desired performance. The aim is to minimise the H^∞ norm of the closed-loop transfer function $T_{\tilde{z}\tilde{w}} = \mathcal{F}_\ell(P_{gen}, C)$, which maps the exogenous inputs $\tilde{w} = \begin{bmatrix} v & w \end{bmatrix}^\top$ to the regulated outputs $\tilde{z} = \begin{bmatrix} z_1 & z_2 \end{bmatrix}^\top$.

The formulation is derived by opening the local positive feedback loop related to the internal model and incorporating the weighting parameters ξ and μ .

The resulting interconnection can be expressed in standard form as follows:

$$\begin{bmatrix} \tilde{z} \\ \tilde{y} \end{bmatrix} = P_{gen} \begin{bmatrix} \tilde{w} \\ u \end{bmatrix}, \quad u = C\tilde{y}, \quad (4.12)$$

where P_{gen} represents the generalised plant and C is the controller to be developed.

The generalised plant P_{gen} includes the original plant P , the low-pass filter F , and the weighting parameters ξ and μ . The parameters ξ and μ are specifically chosen to provide design flexibility and minimise the expression

$$\frac{\gamma_0}{1 - \gamma},$$

where $\gamma_0 = \|T_{ew}\|_\infty$ and $\gamma = \|T_{ba}\|_\infty$, while maintaining the small-gain requirement $\gamma < 1$. The plant is modified compared to (3.15), since for the problem formulation the exogenous signal should be extended from $w = u_g$ to $w = \begin{bmatrix} u_g & i_{ref} \end{bmatrix}$ to include tracking error as an

optimisation objective. Thus, matrices B_1 and D_1 are defined as

$$B_1^e = \begin{bmatrix} \frac{1}{L_1} & 0 \\ -\frac{1}{L_{2g}} & 0 \\ 0 & 0 \end{bmatrix} \quad D_1^e = \begin{bmatrix} 0 & -1 \end{bmatrix}.$$

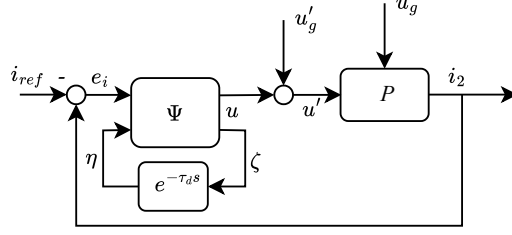
A generalised plant is as follows [18]

$$P_{gen} = \left[\begin{array}{cc|cc|c} A & 0 & 0 & B_1 & B_2 \\ B_F C & A_F & B_F \xi & B_F D_1 & B_F D_2 \\ \hline 0 & C_F & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu \\ C_1 & 0 & \xi & D_1 & D_2 \end{array} \right].$$

The weighting parameters are selected as $\xi = 44$ and $\mu = 0.26$. Employing the MATLAB[®] `hinfsyn` algorithm, the H^∞ controller C , which minimises the H_∞ norm of the transfer matrix from $\tilde{w}$ to $\tilde{z}$, is computed as

$$C(s) = \frac{-2.1818 \times 10^6 (s + 1.311 \times 10^4)(s + 300)(s^2 + 9189s + 4.04 \times 10^8)}{(s + 7.221 \times 10^5)(s^2 + 2.266 \times 10^4 s + 2.567 \times 10^8)(s^2 + 1.718 \times 10^4 s + 3.8 \times 10^8)}.$$

Schemes of the controllers are shown in Figure 4.9. For both techniques $w_\phi = 2\pi \cdot 2550 \text{ rad/s}$. It is important to note that for the current H^∞ repetitive control, this value should not be too high; otherwise, the system will be difficult to stabilise, while the proposed scheme is less restrictive and w_ϕ can be increased. Also, the order of the filter complicates the design. For example, the first-order filter in the repetitive element could be used to make the controller less conservative; however, it would compromise the tracking and disturbance rejection performance.



(a) Repetitive current control with CIM

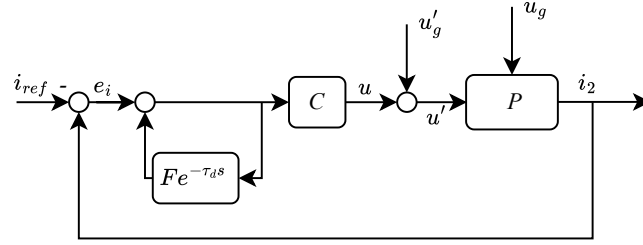
(b) H^∞ repetitive current control

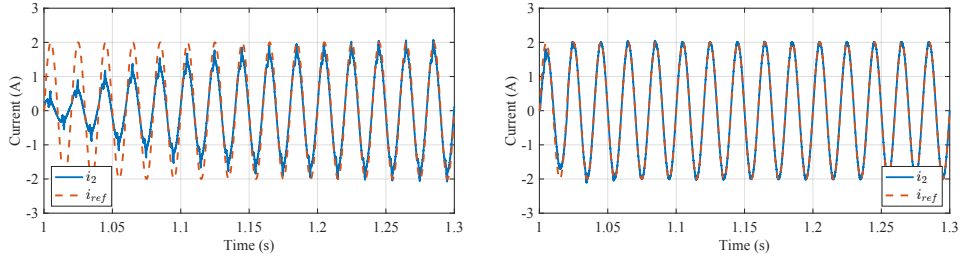
Figure 4.9: Block diagrams of the repetitive current control schemes.

The delay value is not equal to the signal period but is calculated as recommended in [40]

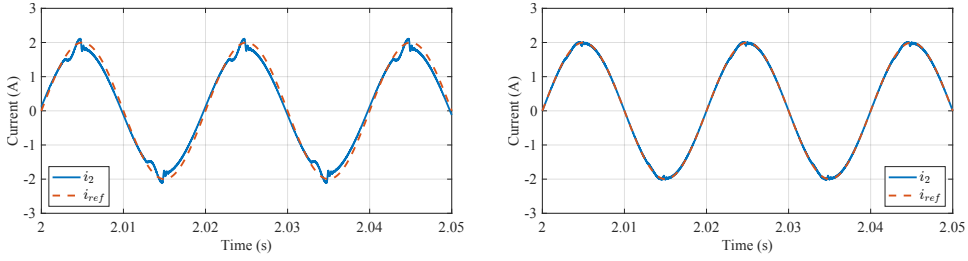
$$\tau_d = \tau - \frac{1}{\omega_\phi}, \quad (4.13)$$

when $w_\phi \tau$ is large, which is normally the case.

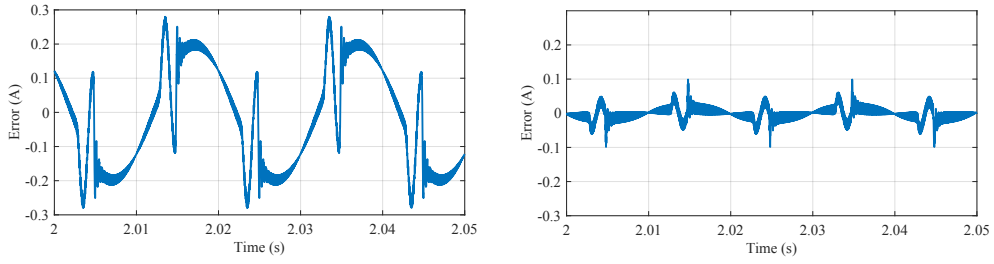
Simulations were done with MATLAB[®]/Simulink[®]. In Figure 4.10, the results are shown when a nonlinear load was connected to the inverter (an uncontrolled converter loaded with an LC filter $L = 150\mu H$, $C = 1000\mu F$, and a resistor $R = 20\Omega$), while in Figure 4.11 simulations with the inverter connected to the grid, which voltage is distorted by harmonics, as in (4.3). Results show better performance in terms of settling time when the current is changed from 0 to 2A (repetitive control with CIM demonstrates much faster transients, avoiding overshoots) and in terms of harmonic rejection for both setups.



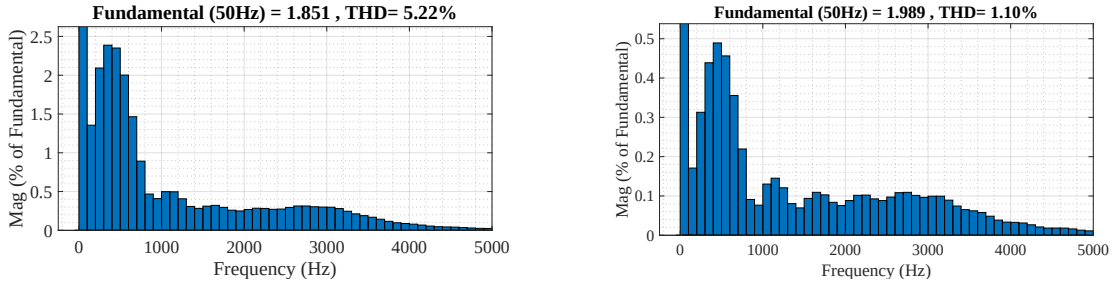
(a) Transient response of the current repetitive controller in the grid-connected mode with local nonlinear load



(b) Grid current i_2 and its reference i_{ref} in the steady-state



(c) Current tracking error e_i in the steady state



(d) Spectra of the grid output current in the steady state

Figure 4.10: Simulation results for repetitive control schemes with nonlinear load: H^∞ repetitive control (left column) and repetitive with CIM (right column).

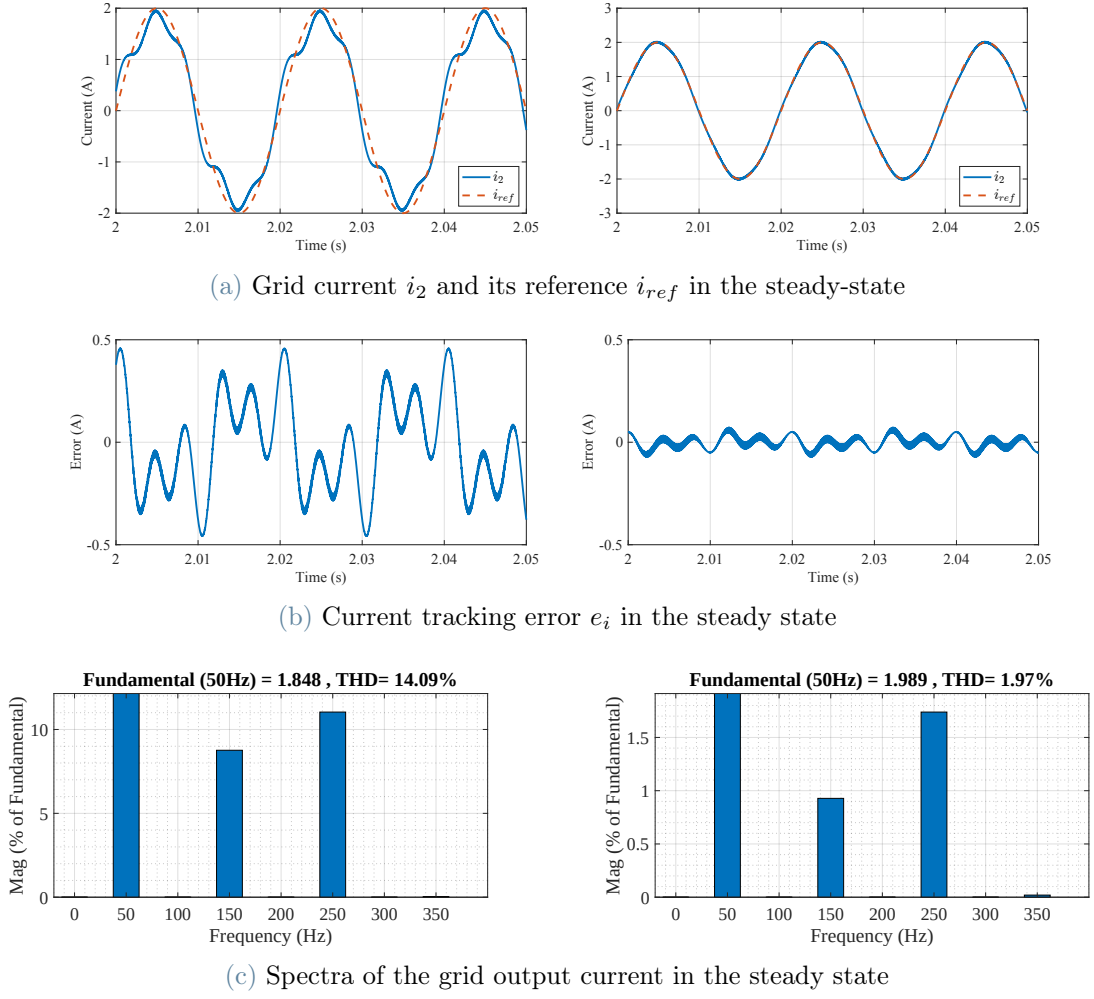


Figure 4.11: Simulation results for control schemes under harmonic distortion: H^∞ repetitive control (left column) and Repetitive with CIM.

A compact quantitative comparison is reported in Table 4.1. Both closed-loop systems are stable, as verified using the Nyquist criterion. Settling time and overshoot are evaluated under nominal operating conditions (no nonlinear load and no harmonic distortion in the grid voltage) for a step change in the amplitude of the current reference (with a fixed waveform and frequency). Robustness is quantified by the minimum Nyquist distance to -1, $d_{\min} = \min_{\omega} |1 + L(j\omega)|$, where L is the open-loop transfer function; smaller values of $d_{\min}$ correspond to a smaller robustness buffer. The results highlight the expected trade-off: the CIM-based repetitive controller achieves strong harmonic rejection at the cost of a slightly reduced robustness buffer compared with the H^∞ -based stabiliser used as a baseline. This can be mitigated by refining the stabiliser design. In particular, since the achieved THD is already well below the H^∞ baseline, there is margin to retune the stabiliser toward increased robustness while maintaining satisfactory harmonic rejection

performance.

	H^∞	CIM
Overshoot, %	< 0.1%	< 0.1%
Settling time, s	0.32	0.02
THD (nonlinear load), %	5.22	1.10
THD (harmonics in grid), %	14.58	1.97
Min. Nyquist distance to -1	0.61	0.53

Table 4.1: Performance and robustness of the repetitive controllers.

4.6. Adaptive repetitive current control with CIM

As was written before, the proposed scheme allows for the introduction of slow adaptation of the delay without compromising the stability of the system. PLL is used for delay estimation, which output frequency f is fed to element Δ (see Figure 4.12), in which the delay element is calculated as

$$\Delta = e^{-(f^{-1} - w_\phi^{-1})s},$$

based on (4.13).

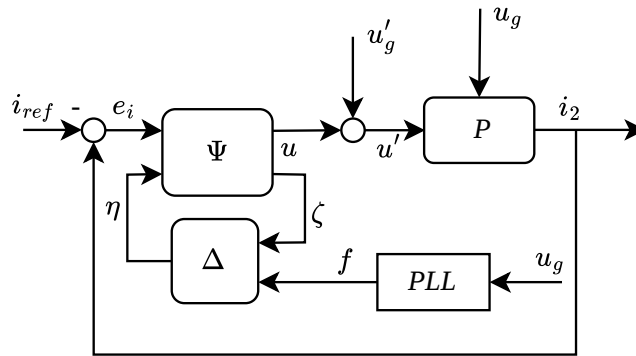
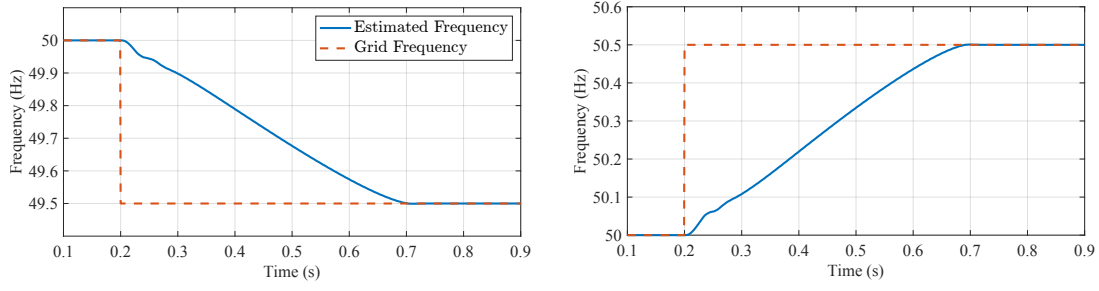


Figure 4.12: Block diagram of the repetitive current control with CIM and delay adaptation.

The stabiliser gain is kept fixed at a value that ensures closed-loop stability. Figure 4.13 shows the PLL frequency estimate after a step change in the grid frequency (Figure 4.13a)

and the resulting grid-current transients (Figures 4.13b–4.13d). It is noticeable that most of the oscillations occur only during the first period after the frequency step change; after this, even during the drift of the estimated frequency that is injected into the delay element, the system remains stable, and the tracking error is small. Figures 4.14–4.17 show the effect of the delay mismatch in the repetitive element and the grid frequency for an inverter connected to the grid with distorted voltage (Figures 4.14–4.15) and when a nonlinear local load is connected to the system (Figures 4.16–4.17). It is noticeable that the output current is more sensitive to the mismatch when the real frequency is higher than the one to which the controller is tuned.



(a) Frequency adaptation

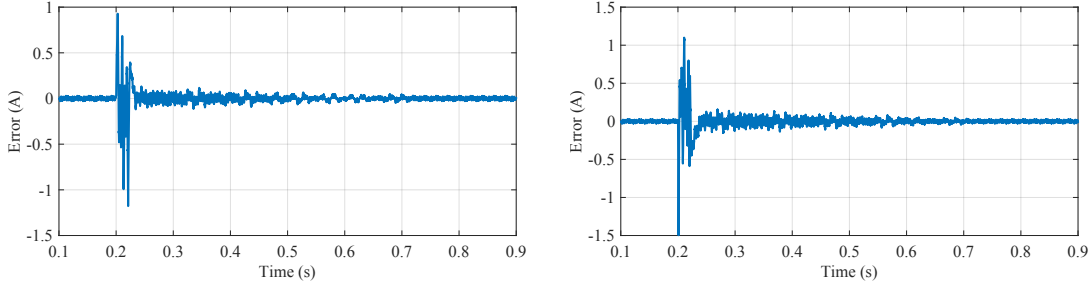
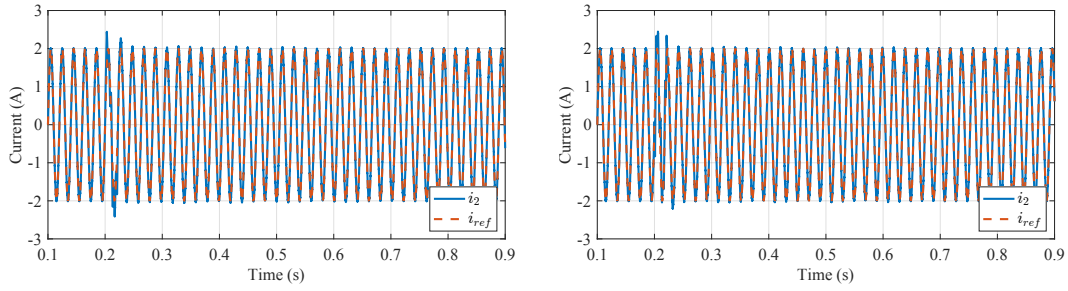
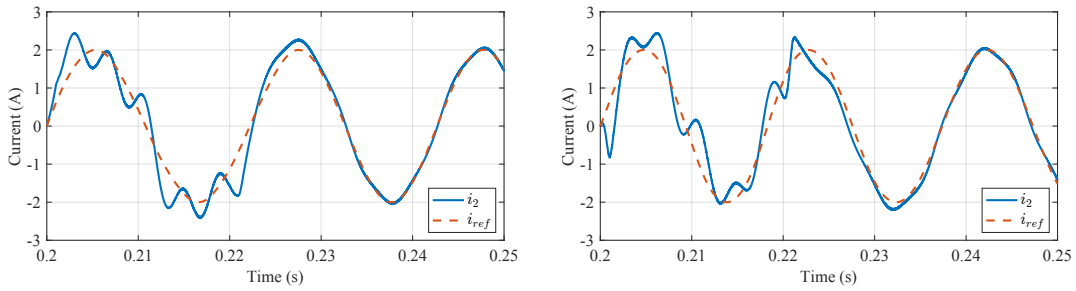
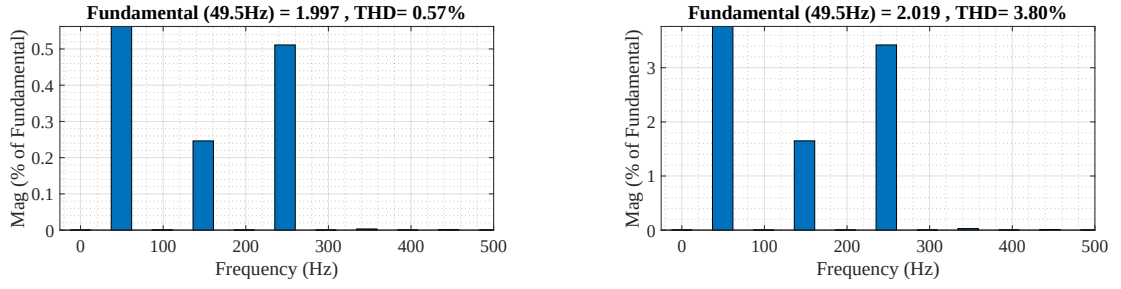
(b) Current tracking error e_i (c) Grid current i_2 and its reference i_{ref} during whole estimation period(d) Grid current i_2 and its reference i_{ref} during first period after frequency change

Figure 4.13: Simulation results for adaptive current repetitive control with CIM: adaptation from 50Hz to 49.5Hz(left column) and 50.5Hz(right column).



(a) Spectra of the grid output current

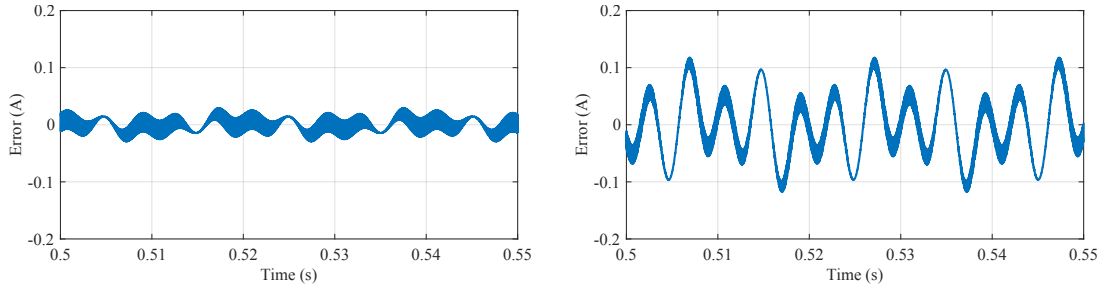
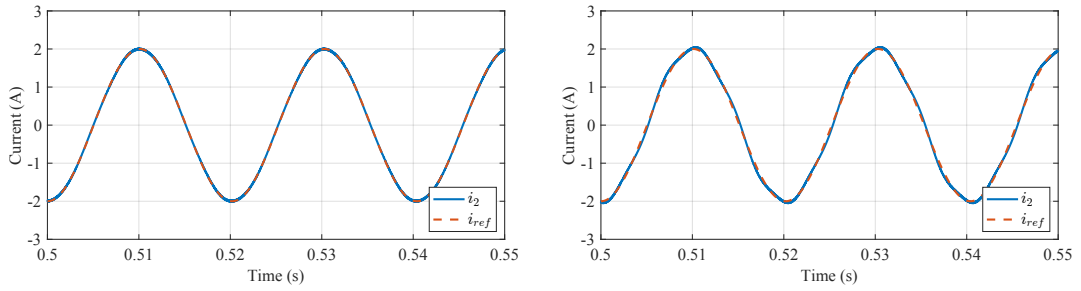
(b) Current tracking error e_i (c) Grid current i_2 and its reference i_{ref}

Figure 4.14: Simulation results for current repetitive control with CIM for inverter connected to the grid with flattened voltage ($f_g = 49.5\text{Hz}$): with adaptive delay (left column) and constant delay set for $f_g = 50\text{Hz}$ (right column).

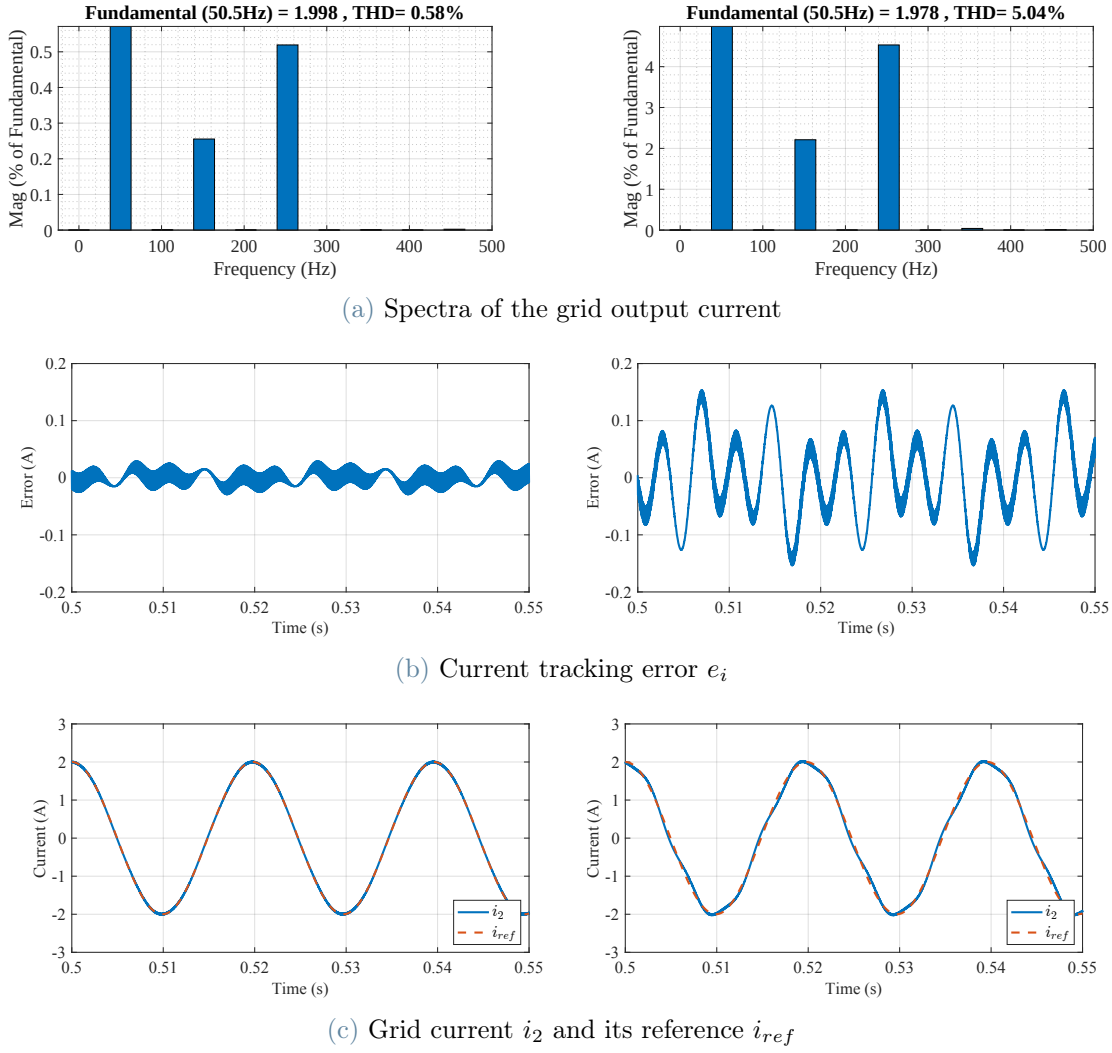
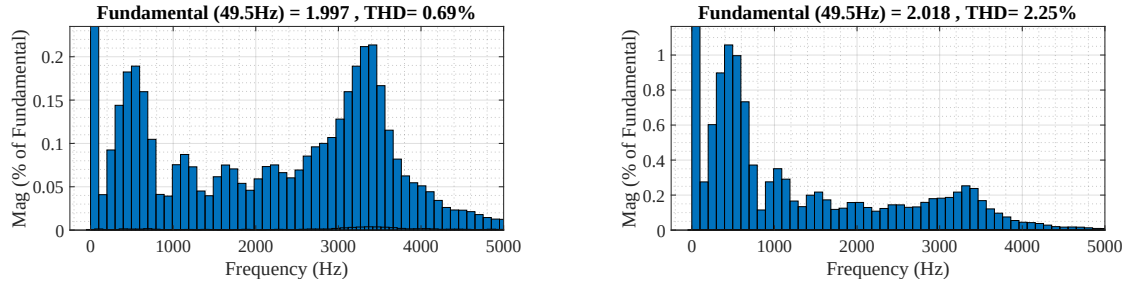


Figure 4.15: Simulation results for current repetitive control with CIM for inverter connected to the grid with flattened voltage ($f_g = 50.5\text{Hz}$): with adaptive delay (left column) and constant delay set for $f_g = 50\text{Hz}$ (right column).



(a) Spectra of the grid output current in the steady state (low frequencies)

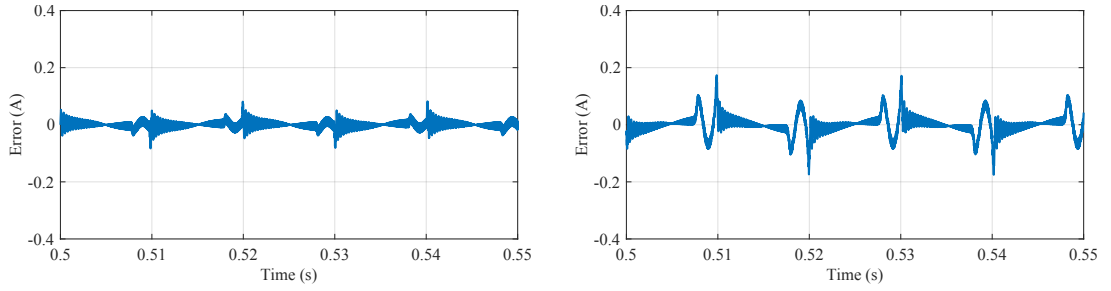
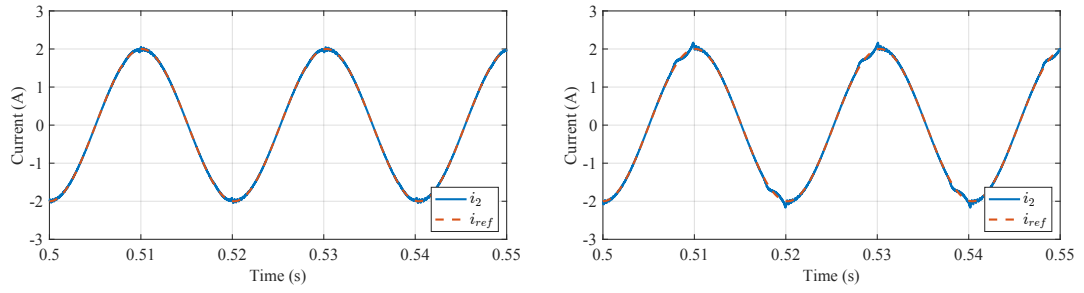
(b) Current tracking error e_i (c) Grid current i_2 and its reference i_{ref}

Figure 4.16: Simulation results for current repetitive control with CIM for the system with nonlinear local load ($f_g = 49.5\text{Hz}$): with adaptive delay (left column) and constant delay set for $f_g = 50\text{Hz}$ (right column).

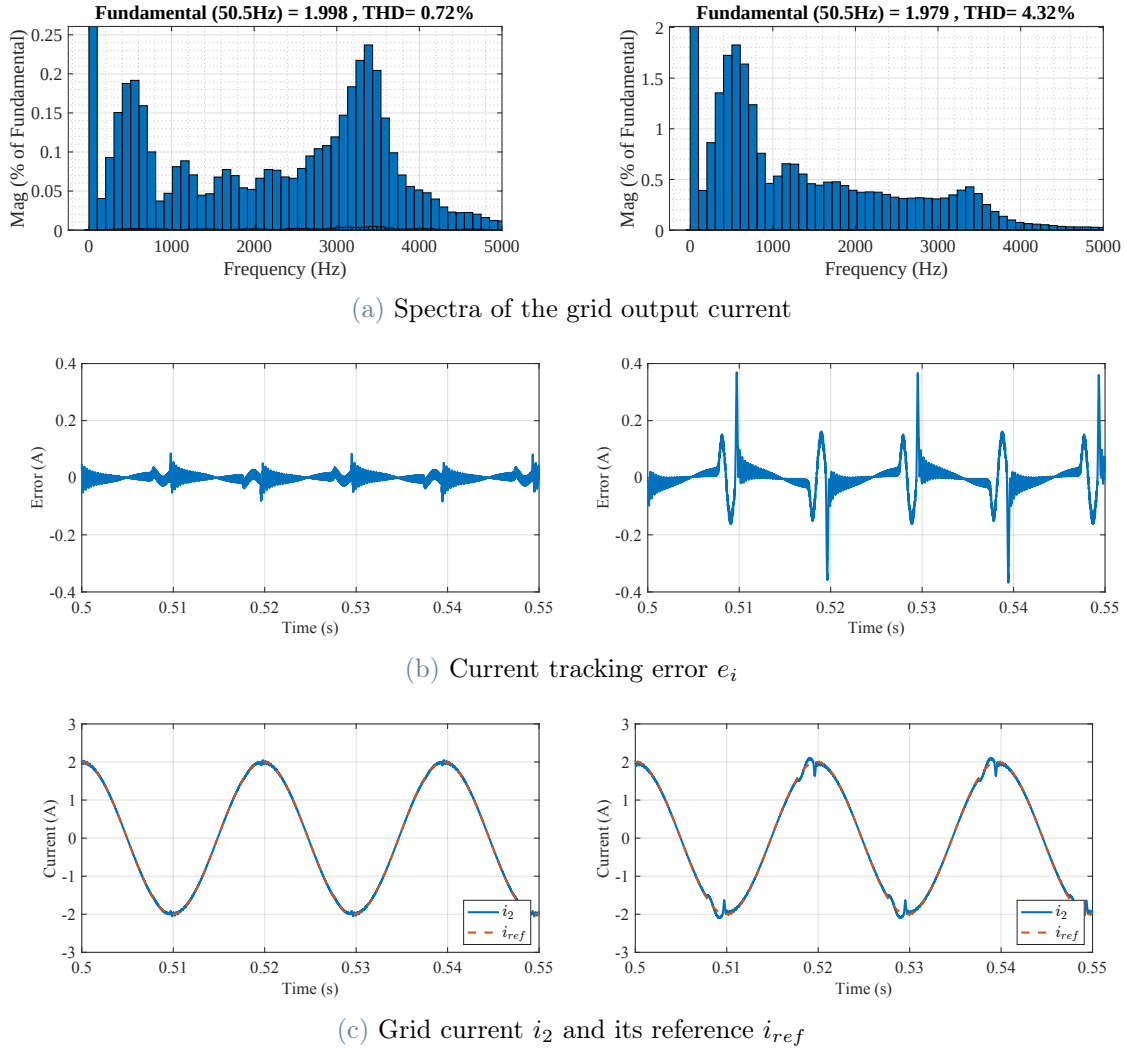


Figure 4.17: Simulation results for current repetitive control with CIM for the system with nonlinear local load ($f_g = 50.5\text{Hz}$): with adaptive delay (left column) and constant delay set for $f_g = 50\text{Hz}$ (right column).

5 | Conclusions and future developments

This thesis examined the output regulation problem in the presence of persistent exogenous signals, with an emphasis on controller structures that (i) comply with the internal model principle (IMP) and (ii) preserve closed-loop stability while internal-model parameters are tuned or adapted under uncertainty. The primary objective was to develop compensation of internal model (CIM) architectures for output regulation, with the aim of facilitating stabiliser design and avoiding stabiliser redesign when internal-model parameters vary over time.

The established framework was then utilised in a benchmark for grid-connected inverters, chosen due to its high power-quality standards and the importance of sinusoidal tracking and harmonic rejection. In this context, the IMP is applied as proportional-resonant (PR) and repetitive controllers. The thesis illustrated that stabiliser gains, which serve to stabilise the equivalent plant, also provide stability to the original plant within the framework of contemporary PR control augmented by internal model compensation. Under certain conditions, these plants have the same order and, in particular cases, may even be identical. Simulation studies demonstrated that CIM-enhanced repetitive control significantly reduces harmonic content in the steady state, resulting in a substantial decrease in total harmonic distortion (THD) in the presented cases.

Lastly, the thesis examined adaptive variants that emerged from the difference between the grid and the controller's internal model frequency. In an idealised scenario with minimal switching distortion, a least-squares adaptation law ensures that frequency estimates converge with high accuracy. Conversely, in a more realistic environment characterised by increased switching distortion, the frequency estimate tends to become noisy and may settle at a biased value. This occurs because the adaptation objective focuses on minimising the current tracking error rather than the error in frequency estimation. A feasible alternative was evaluated in which the frequency estimated by a Phase-Locked Loop (PLL) is provided to the controller. In the scenarios analysed, the steady-state error and Total

Harmonic Distortion (THD) were similar between the pure least-squares method and the PLL-fed approach. This supports the practical recommendation to use the PLL signal when it is available in inverter applications. The thesis demonstrated that the proposed scheme facilitates gradual adjustments to the delay parameter while maintaining stability in the context of repetitive control. Additionally, it revealed that when repetitive control is implemented with CIM, it is more effective at mitigating disturbances caused by harmonic distortion compared to the standard repetitive method, as evidenced by the simulations presented.

The main outcome is a unified design approach for internal-model-based regulation. In this framework, the compensation structures make the controller's dependence on uncertain signal-model parameters clear—ideally in an affine manner—allowing for easier adaptation. At the same time, the stabiliser design remains independent of the internal model. This architectural design is particularly attractive for uses such as grid-connected converters, where (i) periodic external signals hold significant importance, (ii) frequency variations are inevitable, and (iii) practical implementation requirements demand stable and modular controller frameworks. A key practical limitation highlighted in this thesis is that real exogenous signals can fluctuate or change, while traditional IMP focuses on fixed modes. One logical approach is to broaden the adaptive internal-model tuning to explicitly manage transitions among a set of candidate models. Contemporary views on adaptive feedforward and switching internal models provide structured approaches—such as averaging and small-gain arguments, multiple-model frameworks, and estimator-based switching—to ensure internal stability while selecting or adjusting the internal model in the face of uncertainty. The existing CIM architectures could leverage these concepts to demonstrate that frequency steps, variations in multi-harmonic distortion, and uncertainties in the model set remain stable without necessitating a redesign of the stabiliser.

The inverter study primarily focused on controlling single-channel current regulation and adapting to variations in the fundamental frequency. A direct extension entails the simultaneous integration of structured internal models for various harmonics (selective harmonic compensation) and assessing the resulting trade-offs related to complexity, sensitivity to delay, and stability margins. Another extension involves examining multi-output regulation, such as when multiple voltages or currents are controlled simultaneously or when employing three-phase MIMO formulations. In these scenarios, the multiplicity of internal models and the coupling of channels become critically significant. In these circumstances, it is essential to review the output-feedback assumptions. These could be substituted with more flexible conditions or with synthesis methods that ensure adherence to them.

The results from the inverter indicate that switching and sampling can reduce the accuracy of adaptation, as they may produce noisy or biased estimates. This occurs because adaptation focuses on minimising tracking error rather than the error associated with estimating internal-model parameters. Future research ought to incorporate explicit discrete-time models alongside delay-aware design, which should include:

- discrete-time derivations of CIM structures, rather than relying on continuous-time emulation,
- robust synthesis that accommodates delay margins, such as constraints on the placement of resonant peaks,
- estimation laws that take into account the effects of periodic sampling and switching harmonics.

An alternative approach to diminish sensitivity to switching ripple while maintaining convergence in an idealised context is to modify the adaptation objective or incorporate filtering and normalisation techniques.

The adaptation through output feedback relies on internal signals derived from the controller interconnection, as well as on conditions that ensure the equivalent plant remains independent of model parameters. In real converter configurations, measurements are affected by quantisation, delays, and sensor noise. Therefore, it is crucial to develop observer and estimator designs, along with robustification strategies, that maintain boundedness and avoid excitation of unmodeled dynamics. This encompasses:

- an examination of persistent excitation conditions that ensure reliable parameter convergence despite limited harmonic content;
- regularisation techniques in least-squares adaptation, including forgetting factors, projection methods, and dead zones;
- the connection between phase-locked loop (PLL) dynamics and adaptation loops when both are utilised.

The thesis already indicates that using PLL-fed frequency can be beneficial in inverter control; a more thorough analysis could clarify the circumstances under which it is advantageous to depend on PLL frequency, when to integrate it with adaptation, and how to ensure the stability of the combined approach.

The assessment presented is derived from a simulation. An important next step is to conduct experimental validation:

- hardware-in-the-loop validation using realistic PWM, sampling, and measurement systems;
- laboratory inverter setup in grid-following mode with adjustable grid impedance and harmonics;
- stress testing under weak-grid conditions and frequency events (steps/ramps) to assess robustness.

These experiments clarify the constraints involved in implementation, such as computation, choices in discretisation, and considerations for anti-windup and limits. They also provide realistic benchmarks for assessing the effectiveness of pure adaptation compared to parameterisation driven by a phase-locked loop (PLL), particularly in terms of transient quality and harmonic compliance.

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